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(54) **UTILIZATION OF IMMORTALIZED B CELLS TO IDENTIFY SDCBP AS A NOVEL THERAPEUTIC TARGET IN OVARIAN CARCINOMA**

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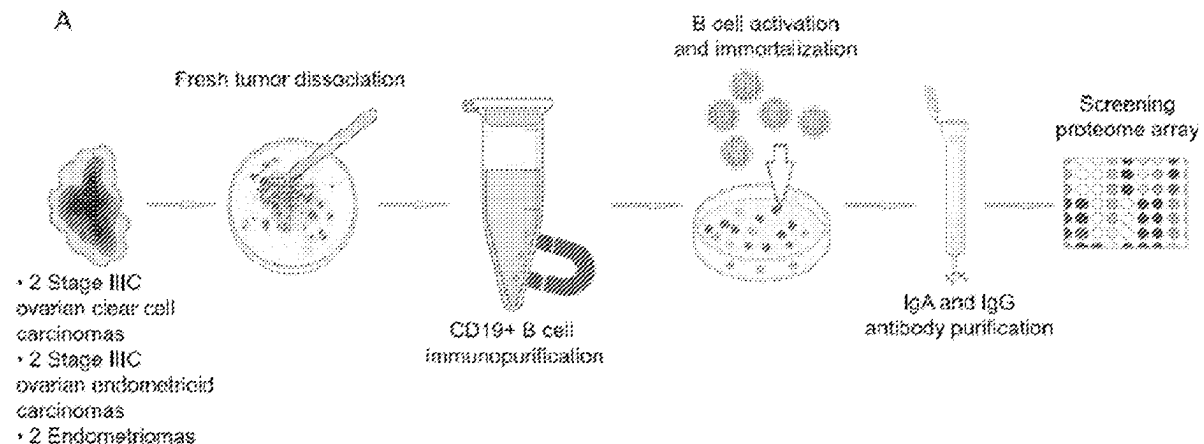
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(57) **ABSTRACT**

Disclosed are anti-syndecan binding protein (SDCBP) binding molecules and methods for using anti-SDCBP in the treatment of cancer, including, but not limited to endometriosis or ovarian cancer.

Specification includes a Sequence Listing.



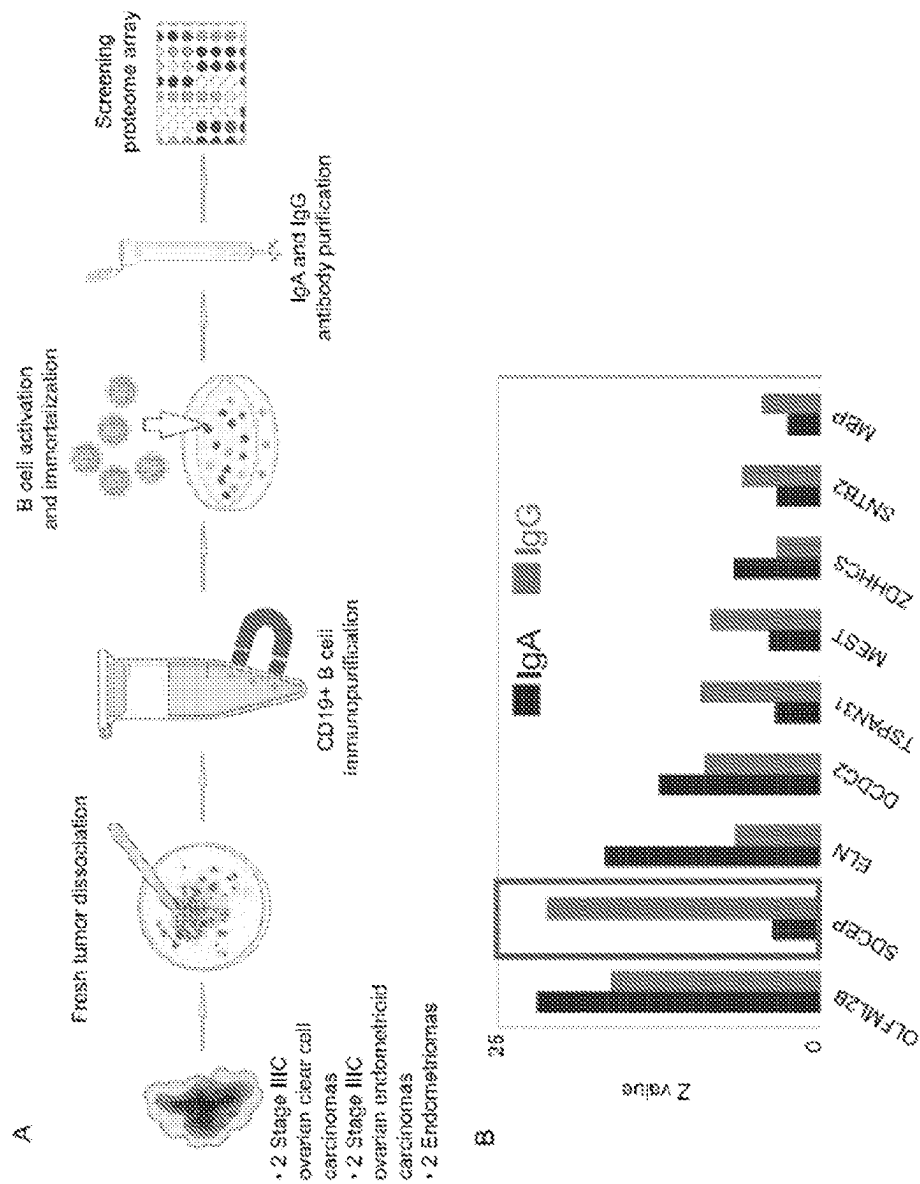


FIG. 1A and FIG. 1B

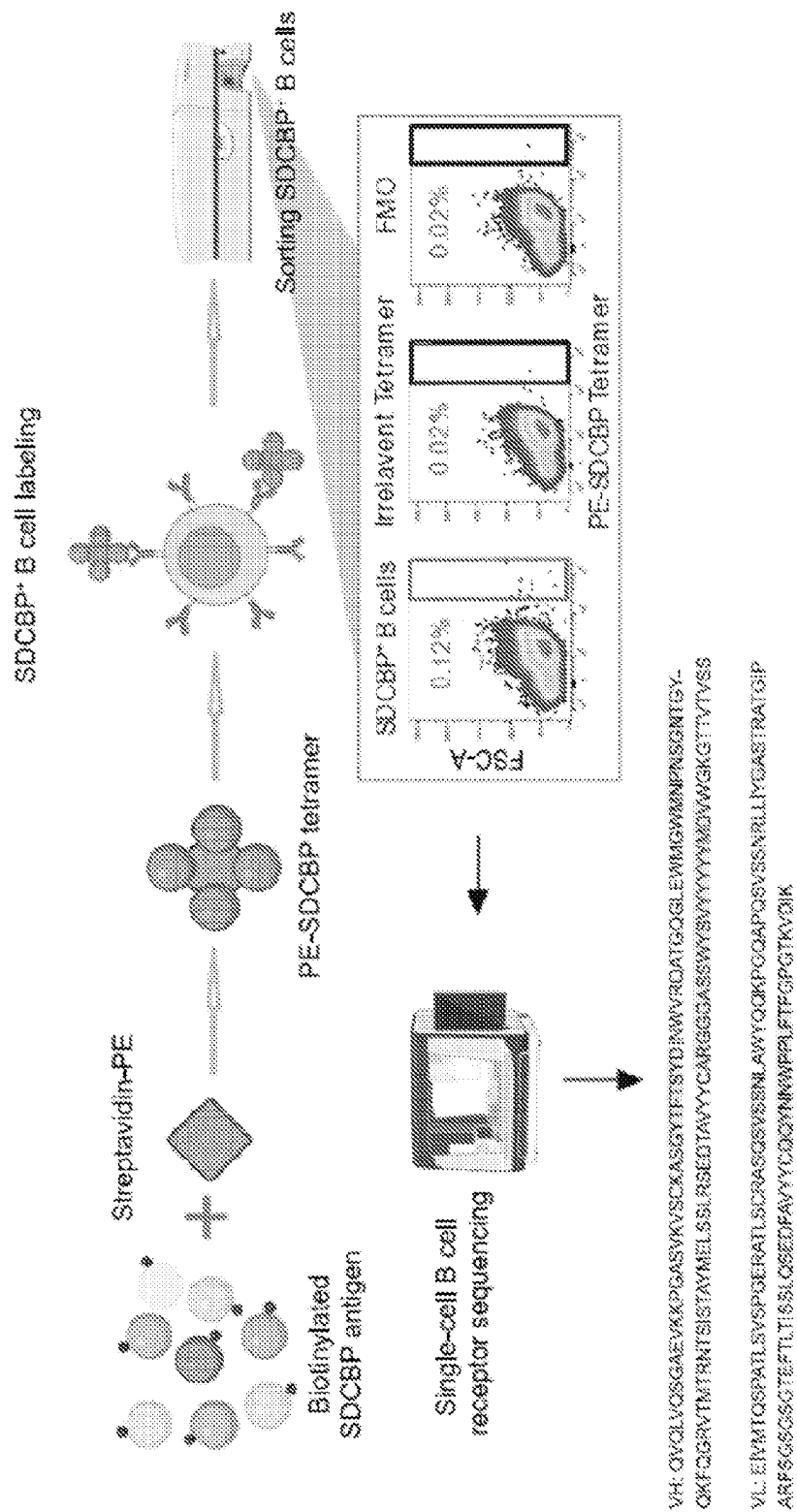


FIG. 2

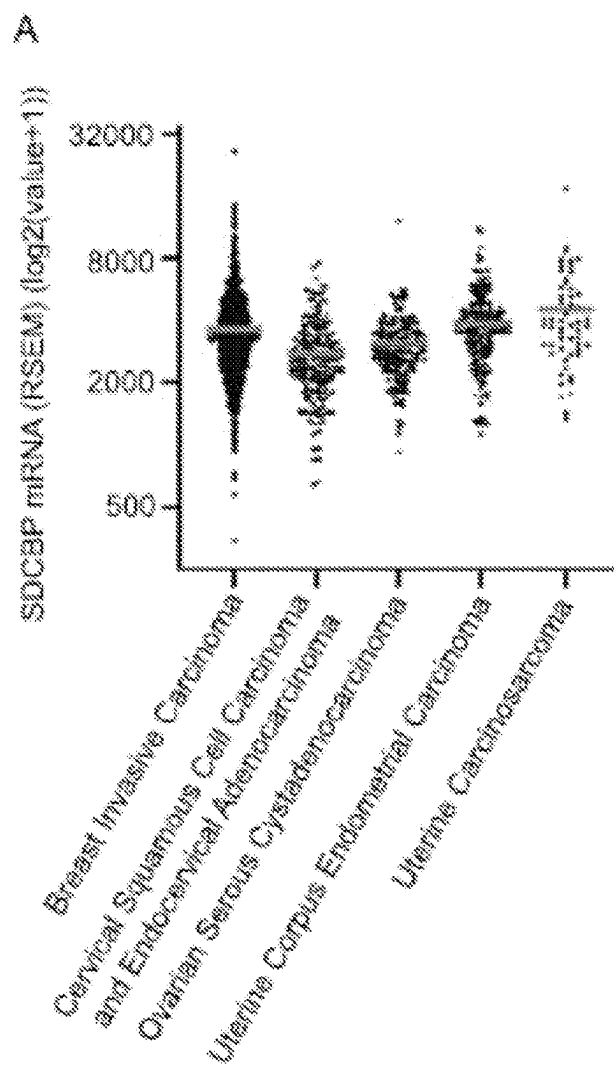


FIG. 3A

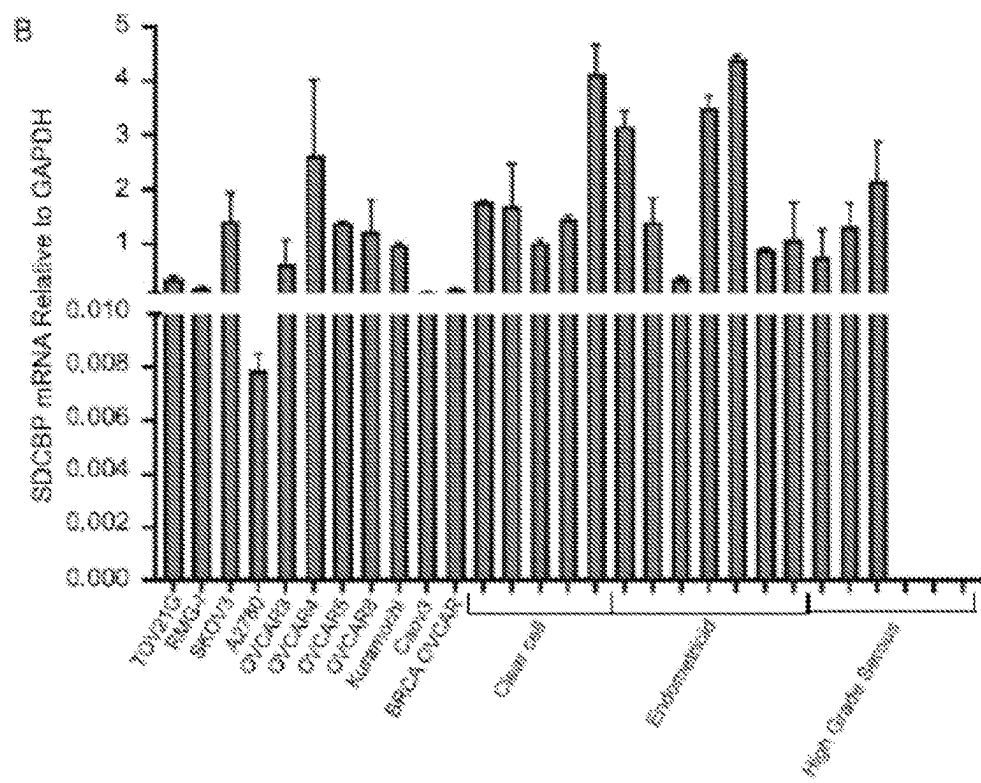


FIG. 3B

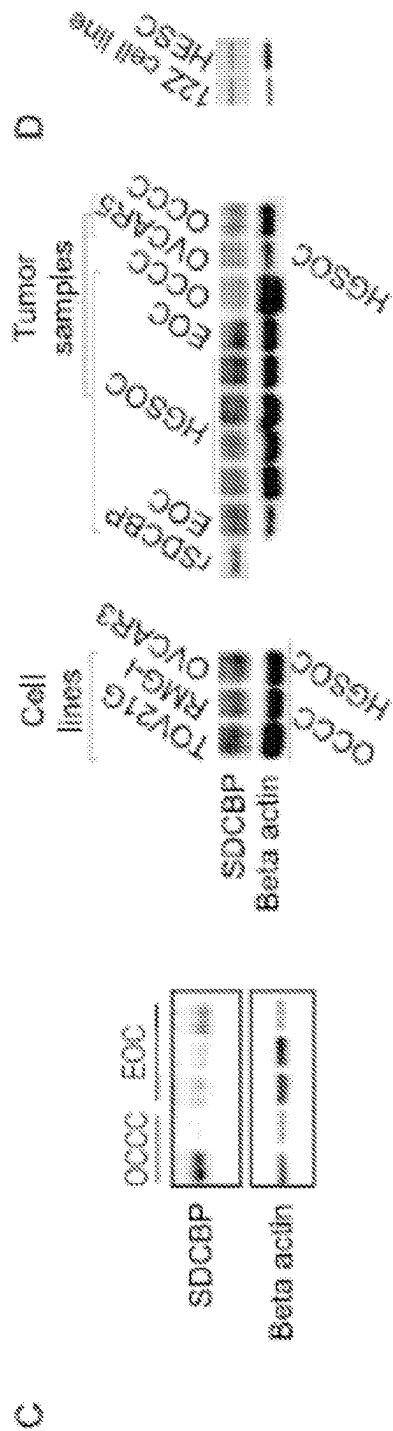


FIG. 3C and FIG. 3D

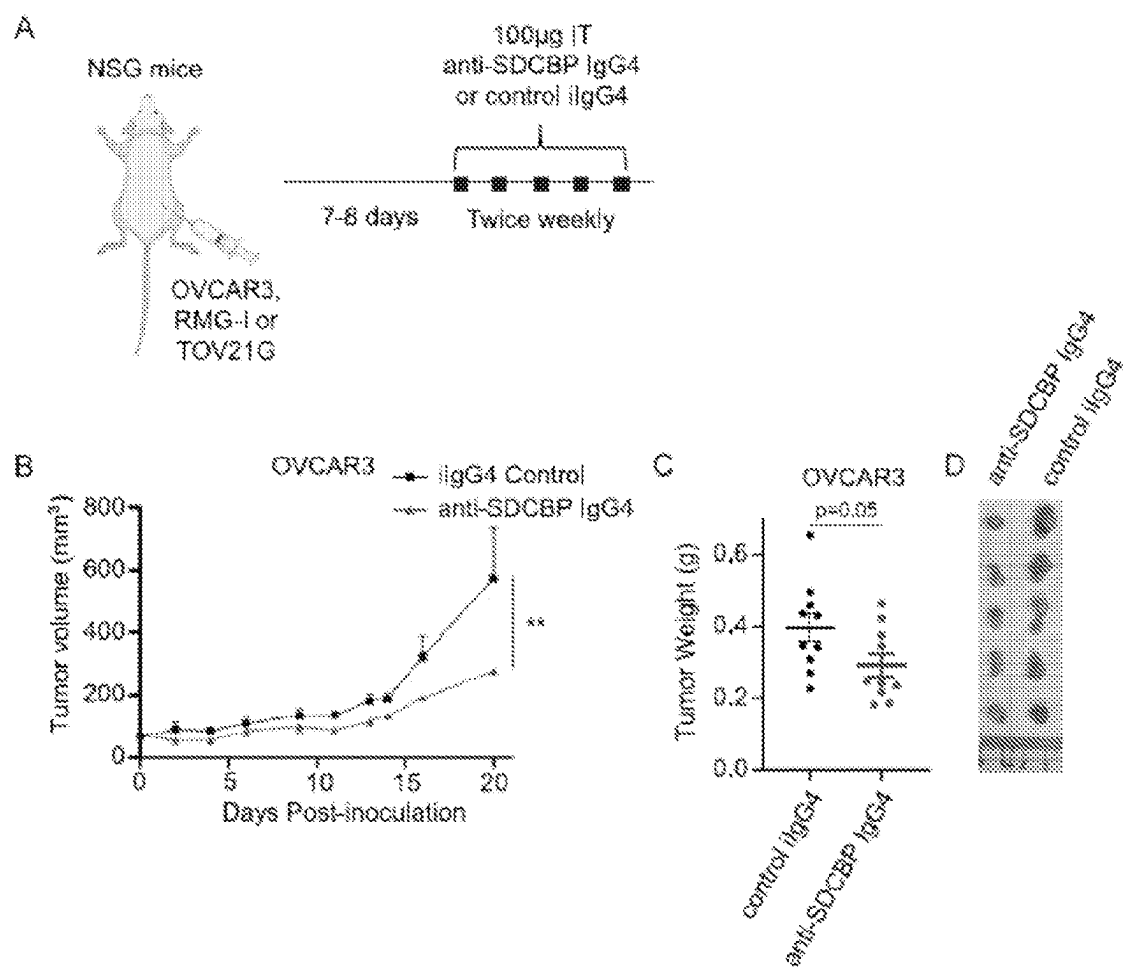


FIG. 4A, FIG. 4B, FIG. 4C, and FIG. 4D

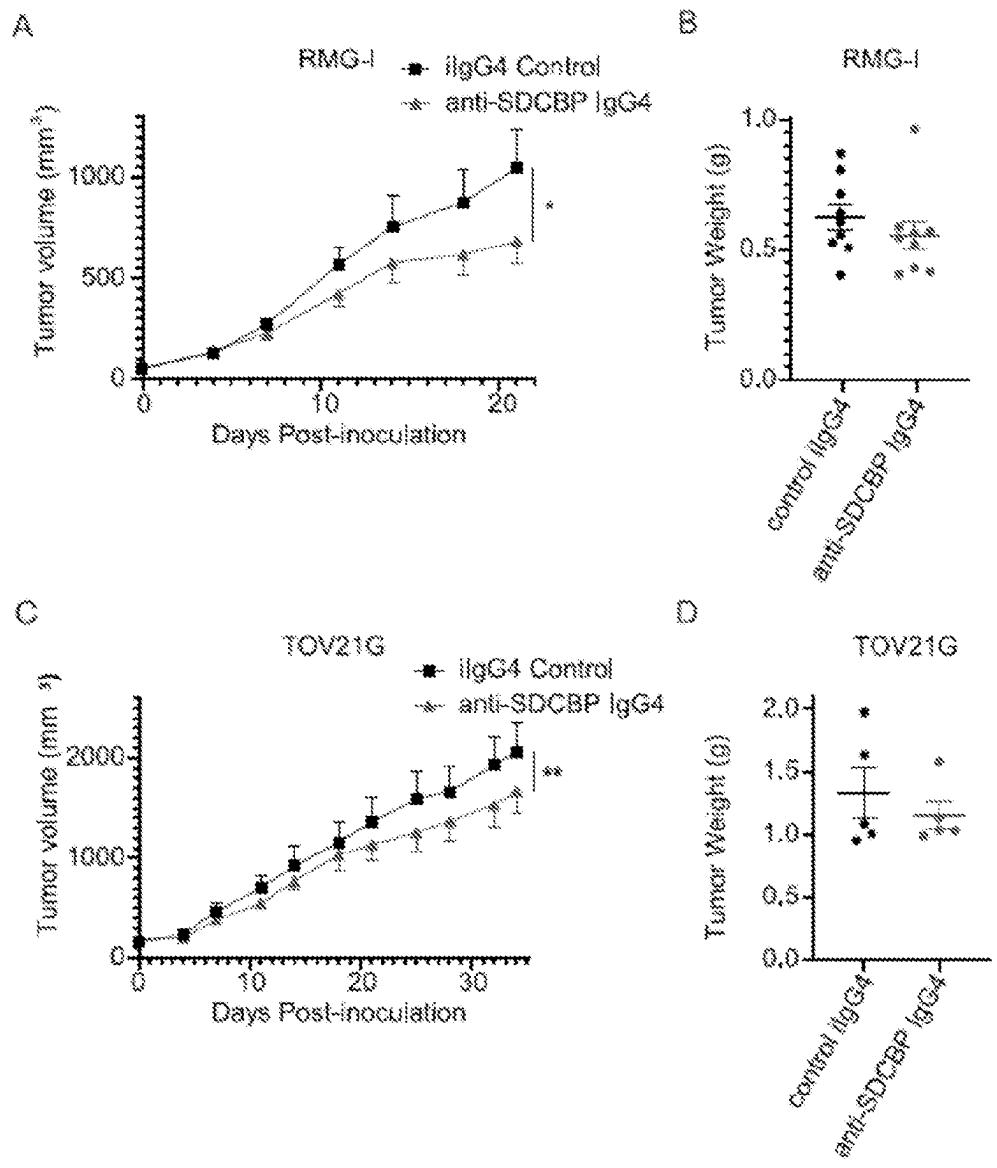
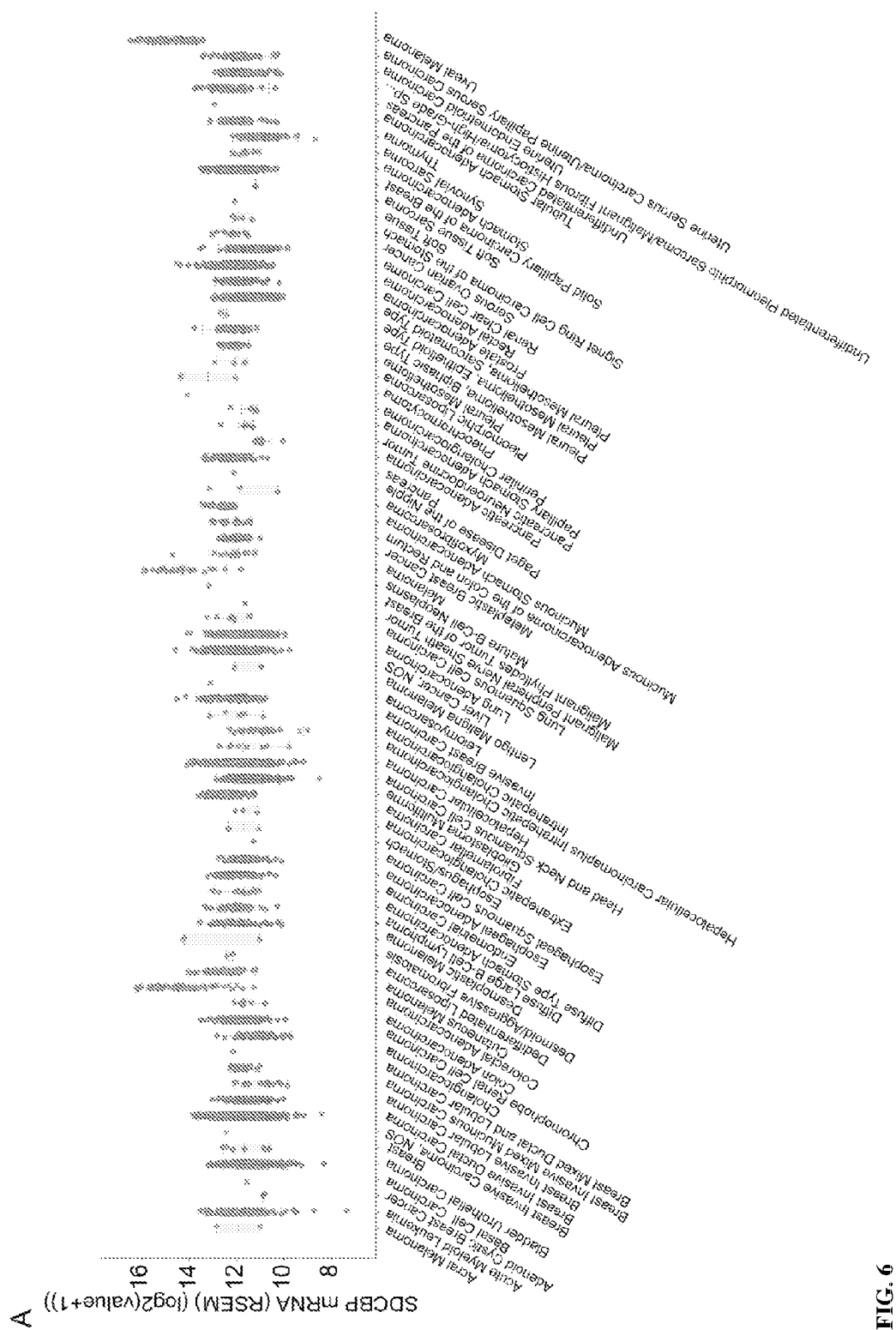


FIG. 5A, FIG. 5B, FIG. 5C, and FIG. 5D



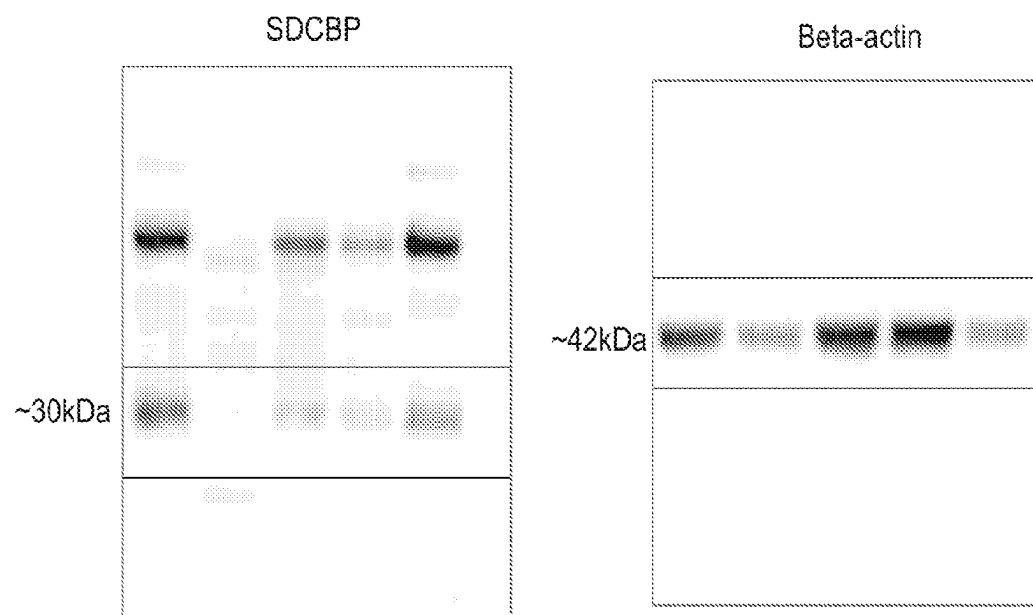


FIG. 7

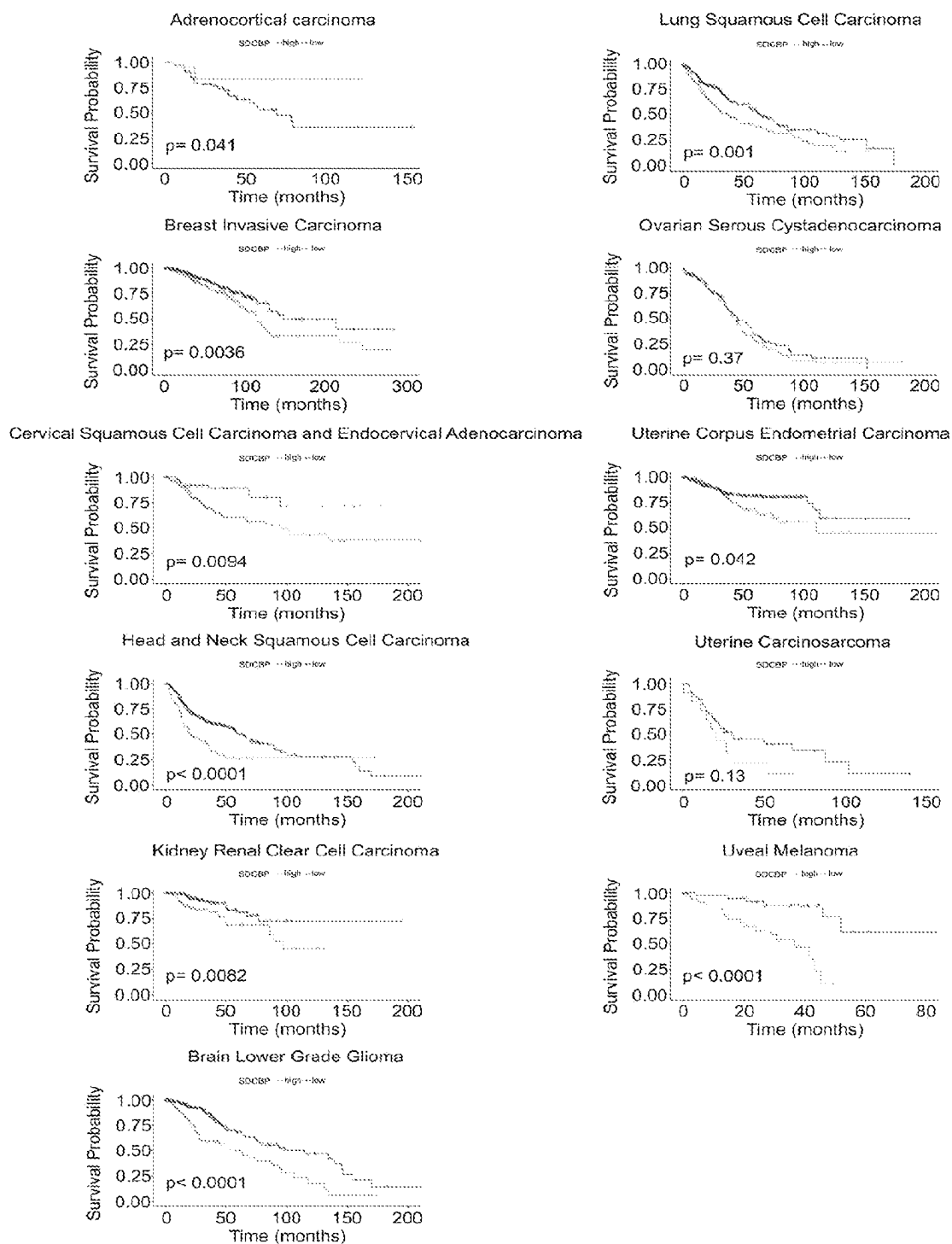


FIG. 8

UTILIZATION OF IMMORTALIZED B CELLS TO IDENTIFY SDCBP AS A NOVEL THERAPEUTIC TARGET IN OVARIAN CARCINOMA

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application claims the benefit of U.S. Provisional Application No. 63/340,186, filed on May 10, 2022, which is incorporated herein by reference in its entirety.

[0002] This invention was made with government support under Grant No. 1K99CA266947-01 and U01CA232758 awarded by the National Institutes of Health. The government has certain rights in the invention.

REFERENCE TO SEQUENCE LISTING

[0003] The sequence listing submitted on May 4, 2023, as an .XML file entitled "10110-372WO1.XML" created on May 4, 2023, and having a file size of 17475 bytes is hereby incorporated by reference pursuant to 37 C.F.R. § 1.52(e)(5).

I. BACKGROUND

[0004] Endometriosis affects approximately 10% of women world wide and is associated with a 2-3 fold increase in the risk of clear cell or endometrioid ovarian cancers. Ovarian endometriosis is associated with a 10-fold increase in the risk of clear cell ovarian cancer and a 5-fold increase in the risk of endometrioid ovarian cancer. Despite for the significant cancer risk, immunotherapies to treat endometriosis or cancers associated with endometriosis are lacking.

[0005] In 2021, there were an estimated 21,410 new cases of ovarian cancer leading to 13,770 deaths. The current standard of care involves aggressive cytoreductive surgery and chemotherapy, which initially elicits a response in more than 80% of cases, but the majority of patients ultimately relapse and develop chemotherapy-resistant disease. The estimated 5-year survival is 49.1% in all cases; the majority of patients are diagnosed at an advanced stage, in which case the 5-year survival drops to 30.1%. There is thus an urgent need for novel therapeutics in this arena.

II. SUMMARY

[0006] Disclosed are methods and compositions related to anti-SDCBP binding molecules for use in treating cancer.

[0007] An anti-syndecan binding protein (SDCBP) binding molecule (such as, for example, an antibody, nanobody, single chain variable fragment (scFv), diabody, or immunotoxin) comprising a heavy chain variable domain complementary determining region (CDR) 1 (CDR1), CDR2, and CDR3 as set forth in SEQ ID NO: 7, SEQ ID NO: 8, and SEQ ID NO: 9, respectively (such as, for example an anti-SDCBP binding molecule comprising a heavy chain variable domain comprising the amino acid sequence as set forth in SEQ ID NO: 6); SEQ ID NO: 7, SEQ ID NO: 8, and SEQ ID NO: 15, respectively; SEQ ID NO: 7, SEQ ID NO: 8, and SEQ ID NO: 16, respectively; or SEQ ID NO: 7, SEQ ID NO: 8, and SEQ ID NO: 17, respectively. In one aspect the heavy chain variable domain comprises the amino acid sequence as set forth in SEQ ID NO: 18.

[0008] Also disclosed herein are anti-SDCBP binding molecules of any preceding aspect, wherein the anti-SDCBP binding molecule further comprises a light chain variable

domain complementary determining region (CDR) 1 (CDR1), CDR2, and CDR3 as set forth in SEQ ID NO: 11, SEQ ID NO: 12, and SEQ ID NO: 13, respectively (such as, for example an anti-SDCBP binding molecule comprising a light chain variable domain comprising the amino acid sequence as set forth in SEQ ID NO: 10) or SEQ ID NO: 11, SEQ ID NO: 12, and SEQ ID NO: 14, respectively. In one aspect the light chain variable domain comprises the amino acid sequence as set forth in SEQ ID NO: 19.

[0009] In one aspect, disclosed herein are methods of treating, inhibiting, reducing, decreasing, ameliorating, and/or preventing a cancer and/or metastasis (such as, for example, ovarian cancer including, but not limited to clear cell carcinoma (CCC), endometrioid carcinoma (EC), high-grade serous ovarian carcinoma (HGSOC), or endometriosis-associated ovarian cancer (EOAC)) or endometriosis in a subject comprising administering to the subject the anti-SDCBP binding molecule of any preceding aspect. For example, disclosed herein are methods treating, inhibiting, reducing, decreasing, ameliorating, and/or preventing a cancer and/or metastasis (such as, for example, ovarian cancer including, but not limited to clear cell carcinoma (CCC), endometrioid carcinoma (EC), high-grade serous ovarian carcinoma (HGSOC), or endometriosis-associated ovarian cancer (EOAC)) or endometriosis in a subject comprising administering to the subject an anti-syndecan binding protein (SDCBP) binding molecule comprising a heavy chain variable domain complementary determining region (CDR) 1 (CDR1), CDR2, and CDR3 as set forth in SEQ ID NO: 7, SEQ ID NO: 8, and SEQ ID NO: 9, respectively (such as, for example an anti-SDCBP binding molecule comprising a heavy chain variable domain comprising the amino acid sequence as set forth in SEQ ID NO: 6); SEQ ID NO: 7, SEQ ID NO: 8, and SEQ ID NO: 15, respectively; SEQ ID NO: 7, SEQ ID NO: 8, and SEQ ID NO: 16, respectively; or SEQ ID NO: 7, SEQ ID NO: 8, and SEQ ID NO: 17, respectively. In one aspect the heavy chain variable domain comprises the amino acid sequence as set forth in SEQ ID NO: 18. In one aspect, the anti-SDCBP binding molecule further comprises a light chain variable domain complementary determining region (CDR) 1 (CDR1), CDR2, and CDR3 as set forth in SEQ ID NO: 11, SEQ ID NO: 12, and SEQ ID NO: 13, respectively (such as, for example an anti-SDCBP binding molecule comprising a light chain variable domain comprising the amino acid sequence as set forth in SEQ ID NO: 10) or SEQ ID NO: 11, SEQ ID NO: 12, and SEQ ID NO: 14, respectively. In one aspect the light chain variable domain comprises the amino acid sequence as set forth in SEQ ID NO: 19.

III. BRIEF DESCRIPTION OF THE DRAWINGS

[0010] The accompanying drawings, which are incorporated in and constitute a part of this specification, illustrate several embodiments and together with the description illustrate the disclosed compositions and methods.

[0011] FIGS. 1A and 1B show tumor infiltrating B cells in clear cell ovarian carcinoma, endometrioid ovarian carcinoma and endometriosis share common antigen targets. FIG. 1A shows a schematic of the optimized protocol for immortalizing B cells from ovarian carcinomas and endometriomas, followed by immunoglobulin purification and characterization with a proteome array. FIG. 1B shows accessible antigens recognized by both IgA and IgG in all clear cell ovarian carcinoma, endometrioid ovarian carcinoma, and

endometrioma specimens. Z value, average Z score (antibody detection signal) of the duplicate spots of a given protein on the HuProt™ array, corrected by the average foreground signal intensity of 2 replicate spots of that protein in the detection channel (background). Z values above 2 were included. Abbreviations: OLFML2B, olfactomedin like 2B; SDCBP, syndecan binding protein; ELN, elastin; DCDC2, doublecortin domain containing 2; TSPAN31, tetraspanin 31; MEST, mesoderm specific transcript; ZDHHC5, zinc finger DHHC-type palmitoyltransferase 2; SNTB2, syntrophin beta 2; MBP, myelin basic protein

[0012] FIG. 2 shows a schematic demonstrating the use of a PE-SDCBP tetramer to sort SDCBP-reactive B cells, which were then subjected to single-cell B cell receptor sequencing with resultant variable heavy (VH) (SEQ ID NO: 6) and variable light (VL) chain (SEQ ID NO: 10) sequences. Complementarity determining regions (CDR) for both the variable heavy chain and variable light chain are highlighted. Abbreviations: SDCBP, syndecan binding protein; PE, phycoerythrin; FMO, fluorescence minus one; FSC-A, forward scatter area

[0013] FIGS. 3A, 3B, 3C, and 3D show SDCBP is expressed in a variety of malignancies, including ovarian clear cell carcinoma, high grade serous carcinoma, and endometrioid carcinoma. FIG. 3A shows the Cancer Genome Atlas data analysis of SDCBP mRNA expression in breast and gynecologic malignancies, measured as transcripts per million mapped reads (TPM). FIG. 3B shows quantitative reverse transcription PCR (RT-qPCR) of SDCBP mRNA expression in ovarian clear cell carcinoma, high grade serous carcinoma, and endometrioid carcinoma cell lines and tissue samples. FIG. 3C shows expression of SDCBP protein in endometrioid ovarian carcinoma (EOC), high grade serous ovarian carcinoma (HGSOC), and ovarian clear cell carcinoma (OCCC) tissue samples; as well as in ovarian clear cell carcinoma and high grade serous ovarian carcinoma cell lines. Positive control, recombinant human SDCBP. FIG. 3D shows expression of SDCBP protein in an endometriosis cell line (12Z) and human endometrial stromal cells (HESC). Abbreviations: SDCBP, syndecan binding protein; GAPDH, glyceraldehyde 3-phosphate dehydrogenase; rSDCBP, recombinant SDCBP

[0014] FIGS. 4A, 4B, 4C, and 4D show that SDCBP can be targeted to treat high grade serous ovarian carcinoma using IgG4 antibodies. FIG. 4A shows a schematic design of in vivo experiments. Anti-SDCBP IgG4 antibody or control IgG4 was injected twice weekly. FIG. 4b shows tumor growth curves, FIG. 4C shows weight, and 4D. volume in OVCAR3 tumor-bearing NSG mice receiving control (irrelevant IgG4 (iIgG4)) or anti-SDCBP IgG4 antibodies. n=10 mice per group. In FIGS. 4B and 4C, growth curves and tumor weights were pooled from two independent experiments. Data are mean±s.e.m. **p≤0.01; Wilcoxon matched-pairs signed rank tests for growth curves and unpaired t test for tumor weights. Abbreviations: NSG, NOD scid gamma; SDCBP, syndecan binding protein; iIgG4, irrelevant IgG4.

[0015] FIGS. 5A, 5B, 5C, and 5D show that SDCBP can be targeted to treat ovarian clear cell carcinomas using IgG4 antibodies in multiple mouse models. FIG. 5A shows tumor growth curves and 5B shows the weight in RMG-I tumor-bearing NSG mice receiving control or anti-SDCBP IgG4 antibodies. n=9 mice per group. FIG. 5C shows tumor growth curves and 5D shows the weight in TOV21G tumor-bearing NSG mice receiving control or anti-SDCBP IgG4

antibodies. n=5 mice per group. This experiment was performed twice with similar results. In FIGS. 5A-5Bb, growth curves and tumor weights were pooled from two independent experiments. Data are mean±s.e.m. *p≤0.05, **p≤0.01; Wilcoxon matched-pairs signed rank tests for growth curves and unpaired t test for tumor weights. Abbreviations: NSG, NOD scid gamma; SDCBP, syndecan binding protein; iIgG4, irrelevant IgG4.

[0016] FIG. 6 shows SDCBP is expressed in a variety of malignancies. The Cancer Genome Atlas data analysis of SDCBP mRNA expression in a variety of malignancies, measured as transcripts per million mapped reads (TPM). Abbreviations: SDCBP, syndecan binding protein.

[0017] FIG. 7 shows that SDCBP is expressed in a variety of malignancies, including ovarian clear cell carcinoma and endometrioid carcinoma. Expression of SDCBP protein in ovarian clear cell carcinoma (n=2) and endometrioid ovarian carcinoma (n=3) tissue samples. Uncropped Western Blots for SDCBP and beta-actin.

[0018] FIG. 8 shows that high SDCBP expression is associated with worse overall survival in breast, cervical, and endometrial cancers, among other cancer types (Cancer Gene Prognosis Atlas, <https://cgpa.moffitt.org/>). Abbreviations: SDCBP, syndecan binding protein.

IV. DETAILED DESCRIPTION

[0019] Before the present compounds, compositions, articles, devices, and/or methods are disclosed and described, it is to be understood that they are not limited to specific synthetic methods or specific recombinant biotechnology methods unless otherwise specified, or to particular reagents unless otherwise specified, as such may, of course, vary. It is also to be understood that the terminology used herein is for the purpose of describing particular embodiments only and is not intended to be limiting.

A. DEFINITIONS

[0020] As used in the specification and the appended claims, the singular forms “a,” “an” and “the” include plural referents unless the context clearly dictates otherwise. Thus, for example, reference to “a pharmaceutical carrier” includes mixtures of two or more such carriers, and the like.

[0021] Ranges can be expressed herein as from “about” one particular value, and/or to “about” another particular value. When such a range is expressed, another embodiment includes from the one particular value and/or to the other particular value. Similarly, when values are expressed as approximations, by use of the antecedent “about,” it will be understood that the particular value forms another embodiment. It will be further understood that the endpoints of each of the ranges are significant both in relation to the other endpoint, and independently of the other endpoint. It is also understood that there are a number of values disclosed herein, and that each value is also herein disclosed as “about” that particular value in addition to the value itself. For example, if the value “10” is disclosed, then “about 10” is also disclosed. It is also understood that when a value is disclosed that “less than or equal to” the value, “greater than or equal to the value” and possible ranges between values are also disclosed, as appropriately understood by the skilled artisan. For example, if the value “10” is disclosed the “less than or equal to 10” as well as “greater than or equal to 10” is also disclosed. It is also understood that the throughout the

application, data is provided in a number of different formats, and that this data, represents endpoints and starting points, and ranges for any combination of the data points. For example, if a particular data point “10” and a particular data point 15 are disclosed, it is understood that greater than, greater than or equal to, less than, less than or equal to, and equal to 10 and 15 are considered disclosed as well as between 10 and 15. It is also understood that each unit between two particular units are also disclosed. For example, if 10 and 15 are disclosed, then 11, 12, 13, and 14 are also disclosed.

[0022] In this specification and in the claims which follow, reference will be made to a number of terms which shall be defined to have the following meanings:

[0023] “Optional” or “optionally” means that the subsequently described event or circumstance may or may not occur, and that the description includes instances where said event or circumstance occurs and instances where it does not.

[0024] An “increase” can refer to any change that results in a greater amount of a symptom, disease, composition, condition or activity. An increase can be any individual, median, or average increase in a condition, symptom, activity, composition in a statistically significant amount. Thus, the increase can be a 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 15, 20, 25, 30, 35, 40, 45, 50, 55, 60, 65, 70, 75, 80, 85, 90, 95, or 100% increase so long as the increase is statistically significant.

[0025] A “decrease” can refer to any change that results in a smaller amount of a symptom, disease, composition, condition, or activity. A substance is also understood to decrease the genetic output of a gene when the genetic output of the gene product with the substance is less relative to the output of the gene product without the substance. Also for example, a decrease can be a change in the symptoms of a disorder such that the symptoms are less than previously observed. A decrease can be any individual, median, or average decrease in a condition, symptom, activity, composition in a statistically significant amount. Thus, the decrease can be a 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 15, 20, 25, 30, 35, 40, 45, 50, 55, 60, 65, 70, 75, 80, 85, 90, 95, or 100% decrease so long as the decrease is statistically significant.

[0026] “Inhibit,” “inhibiting,” and “inhibition” mean to decrease an activity, response, condition, disease, or other biological parameter. This can include but is not limited to the complete ablation of the activity, response, condition, or disease. This may also include, for example, a 10% reduction in the activity, response, condition, or disease as compared to the native or control level. Thus, the reduction can be a 10, 20, 30, 40, 50, 60, 70, 80, 90, 100%, or any amount of reduction in between as compared to native or control levels.

[0027] By “reduce” or other forms of the word, such as “reducing” or “reduction,” is meant lowering of an event or characteristic (e.g., tumor growth). It is understood that this is typically in relation to some standard or expected value, in other words it is relative, but that it is not always necessary for the standard or relative value to be referred to. For example, “reduces tumor growth” means reducing the rate of growth of a tumor relative to a standard or a control.

[0028] By “prevent” or other forms of the word, such as “preventing” or “prevention,” is meant to stop a particular event or characteristic, to stabilize or delay the development or progression of a particular event or characteristic, or to minimize the chances that a particular event or characteristic

will occur. Prevent does not require comparison to a control as it is typically more absolute than, for example, reduce. As used herein, something could be reduced but not prevented, but something that is reduced could also be prevented. Likewise, something could be prevented but not reduced, but something that is prevented could also be reduced. It is understood that where reduce or prevent are used, unless specifically indicated otherwise, the use of the other word is also expressly disclosed.

[0029] The term “subject” refers to any individual who is the target of administration or treatment. The subject can be a vertebrate, for example, a mammal. In one aspect, the subject can be human, non-human primate, bovine, equine, porcine, canine, or feline. The subject can also be a guinea pig, rat, hamster, rabbit, mouse, or mole. Thus, the subject can be a human or veterinary patient. The term “patient” refers to a subject under the treatment of a clinician, e.g., physician.

[0030] The term “therapeutically effective” refers to the amount of the composition used is of sufficient quantity to ameliorate one or more causes or symptoms of a disease or disorder. Such amelioration only requires a reduction or alteration, not necessarily elimination.

[0031] The term “treatment” refers to the medical management of a patient with the intent to cure, ameliorate, stabilize, or prevent a disease, pathological condition, or disorder. This term includes active treatment, that is, treatment directed specifically toward the improvement of a disease, pathological condition, or disorder, and also includes causal treatment, that is, treatment directed toward removal of the cause of the associated disease, pathological condition, or disorder. In addition, this term includes palliative treatment, that is, treatment designed for the relief of symptoms rather than the curing of the disease, pathological condition, or disorder; preventative treatment, that is, treatment directed to minimizing or partially or completely inhibiting the development of the associated disease, pathological condition, or disorder; and supportive treatment, that is, treatment employed to supplement another specific therapy directed toward the improvement of the associated disease, pathological condition, or disorder.

[0032] “Biocompatible” generally refers to a material and any metabolites or degradation products thereof that are generally non-toxic to the recipient and do not cause significant adverse effects to the subject.

[0033] “Comprising” is intended to mean that the compositions, methods, etc. include the recited elements, but do not exclude others. “Consisting essentially of” when used to define compositions and methods, shall mean including the recited elements, but excluding other elements of any essential significance to the combination. Thus, a composition consisting essentially of the elements as defined herein would not exclude trace contaminants from the isolation and purification method and pharmaceutically acceptable carriers, such as phosphate buffered saline, preservatives, and the like. “Consisting of” shall mean excluding more than trace elements of other ingredients and substantial method steps for administering the compositions provided and/or claimed in this disclosure. Embodiments defined by each of these transition terms are within the scope of this disclosure.

[0034] A “control” is an alternative subject or sample used in an experiment for comparison purposes. A control can be “positive” or “negative.”

[0035] “Effective amount” of an agent refers to a sufficient amount of an agent to provide a desired effect. The amount of agent that is “effective” will vary from subject to subject, depending on many factors such as the age and general condition of the subject, the particular agent or agents, and the like. Thus, it is not always possible to specify a quantified “effective amount.” However, an appropriate “effective amount” in any subject case may be determined by one of ordinary skill in the art using routine experimentation. Also, as used herein, and unless specifically stated otherwise, an “effective amount” of an agent can also refer to an amount covering both therapeutically effective amounts and prophylactically effective amounts. An “effective amount” of an agent necessary to achieve a therapeutic effect may vary according to factors such as the age, sex, and weight of the subject. Dosage regimens can be adjusted to provide the optimum therapeutic response. For example, several divided doses may be administered daily or the dose may be proportionally reduced as indicated by the exigencies of the therapeutic situation.

[0036] A “pharmaceutically acceptable” component can refer to a component that is not biologically or otherwise undesirable, i.e., the component may be incorporated into a pharmaceutical formulation provided by the disclosure and administered to a subject as described herein without causing significant undesirable biological effects or interacting in a deleterious manner with any of the other components of the formulation in which it is contained. When used in reference to administration to a human, the term generally implies the component has met the required standards of toxicological and manufacturing testing or that it is included on the Inactive Ingredient Guide prepared by the U.S. Food and Drug Administration.

[0037] “Pharmaceutically acceptable carrier” (sometimes referred to as a “carrier”) means a carrier or excipient that is useful in preparing a pharmaceutical or therapeutic composition that is generally safe and non-toxic and includes a carrier that is acceptable for veterinary and/or human pharmaceutical or therapeutic use. The terms “carrier” or “pharmaceutically acceptable carrier” can include, but are not limited to, phosphate buffered saline solution, water, emulsions (such as an oil/water or water/oil emulsion) and/or various types of wetting agents. As used herein, the term “carrier” encompasses, but is not limited to, any excipient, diluent, filler, salt, buffer, stabilizer, solubilizer, lipid, stabilizer, or other material well known in the art for use in pharmaceutical formulations and as described further herein.

[0038] “Pharmacologically active” (or simply “active”), as in a “pharmacologically active” derivative or analog, can refer to a derivative or analog (e.g., a salt, ester, amide, conjugate, metabolite, isomer, fragment, etc.) having the same type of pharmacological activity as the parent compound and approximately equivalent in degree.

[0039] “Therapeutic agent” refers to any composition that has a beneficial biological effect. Beneficial biological effects include both therapeutic effects, e.g., treatment of a disorder or other undesirable physiological condition, and prophylactic effects, e.g., prevention of a disorder or other undesirable physiological condition (e.g., a non-immunogenic cancer). The terms also encompass pharmaceutically acceptable, pharmacologically active derivatives of beneficial agents specifically mentioned herein, including, but not limited to, salts, esters, amides, proagents, active metabolites, isomers, fragments, analogs, and the like. When the

terms “therapeutic agent” is used, then, or when a particular agent is specifically identified, it is to be understood that the term includes the agent per se as well as pharmaceutically acceptable, pharmacologically active salts, esters, amides, proagents, conjugates, active metabolites, isomers, fragments, analogs, etc.

[0040] “Therapeutically effective amount” or “therapeutically effective dose” of a composition (e.g. a composition comprising an agent) refers to an amount that is effective to achieve a desired therapeutic result. In some embodiments, a desired therapeutic result is the control of type I diabetes. In some embodiments, a desired therapeutic result is the control of obesity. Therapeutically effective amounts of a given therapeutic agent will typically vary with respect to factors such as the type and severity of the disorder or disease being treated and the age, gender, and weight of the subject. The term can also refer to an amount of a therapeutic agent, or a rate of delivery of a therapeutic agent (e.g., amount over time), effective to facilitate a desired therapeutic effect, such as pain relief. The precise desired therapeutic effect will vary according to the condition to be treated, the tolerance of the subject, the agent and/or agent formulation to be administered (e.g., the potency of the therapeutic agent, the concentration of agent in the formulation, and the like), and a variety of other factors that are appreciated by those of ordinary skill in the art. In some instances, a desired biological or medical response is achieved following administration of multiple dosages of the composition to the subject over a period of days, weeks, or years.

[0041] Throughout this application, various publications are referenced. The disclosures of these publications in their entireties are hereby incorporated by reference into this application in order to more fully describe the state of the art to which this pertains. The references disclosed are also individually and specifically incorporated by reference herein for the material contained in them that is discussed in the sentence in which the reference is relied upon.

B. COMPOSITIONS

[0042] Disclosed are the components to be used to prepare the disclosed compositions as well as the compositions themselves to be used within the methods disclosed herein. These and other materials are disclosed herein, and it is understood that when combinations, subsets, interactions, groups, etc. of these materials are disclosed that while specific reference of each various individual and collective combinations and permutation of these compounds may not be explicitly disclosed, each is specifically contemplated and described herein. For example, if a particular anti-SDCBP binding molecule is disclosed and discussed and a number of modifications that can be made to a number of molecules including the anti-SDCBP binding molecule are discussed, specifically contemplated is each and every combination and permutation of anti-SDCBP binding molecule and the modifications that are possible unless specifically indicated to the contrary. Thus, if a class of molecules A, B, and C are disclosed as well as a class of molecules D, E, and F and an example of a combination molecule, A-D is disclosed, then even if each is not individually recited each is individually and collectively contemplated meaning combinations, A-E, A-F, B-D, B-E, B-F, C-D, C-E, and C-F are considered disclosed. Likewise, any subset or combination of these is also disclosed. Thus, for example, the sub-group of A-E, B-F, and C-E would be considered disclosed. This concept

applies to all aspects of this application including, but not limited to, steps in methods of making and using the disclosed compositions. Thus, if there are a variety of additional steps that can be performed it is understood that each of these additional steps can be performed with any specific embodiment or combination of embodiments of the disclosed methods.

[0043] While immunotherapies such as immune checkpoint blockade are changing the trajectory of various cancers such as melanoma, the results in ovarian cancer have been disappointing. Interestingly, tumor-infiltrating B lymphocytes are present in most ovarian cancers and have been associated with improved patient outcomes. However, limited research has focused on humoral response in endometriosis-associated ovarian cancers or endometriosis.

[0044] Endometriosis is a condition in which endometrial tissue is present outside of the uterine cavity, which occurs in approximately 10% of women. Endometriosis is associated with a two- to three-fold increase in a woman's risk of developing clear cell or endometrioid ovarian cancers, and ovarian endometriosis has been associated with as high as a ten-fold increased risk of clear cell and five-fold risk of endometrioid ovarian cancer. Recent studies have concluded that endometriosis is a precursor lesion to these endometriosis-associated ovarian cancers (EAOC), with corresponding somatic mutations identified in both.

[0045] Therefore, we hypothesized that antitumor humoral responses generated by tumor-infiltrating B lymphocytes in ovarian cancer and B lymphocytes in the microenvironment of endometriosis target specific antigens that can be used to identify novel, targetable antigen domains that can exert a protective effect on progression of established ovarian carcinomas.

[0046] In one aspect disclosed herein are anti-syndecan binding protein (SDCBP) binding molecules (such as, for example, an antibody, nanobody, single chain variable fragment (scFv), diabody, or immunotoxin) comprising a heavy chain variable domain complementary determining region (CDR) 1 (CDR1), CDR2, and CDR3 as set forth in SEQ ID NO: 7, SEQ ID NO: 8, and SEQ ID NO: 9, respectively (such as, for example an anti-SDCBP binding molecule comprising a heavy chain variable domain comprising the amino acid sequence as set forth in SEQ ID NO: 6); SEQ ID NO: 7, SEQ ID NO: 8, and SEQ ID NO: 15, respectively; SEQ ID NO: 7, SEQ ID NO: 8, and SEQ ID NO: 16, respectively; or SEQ ID NO: 7, SEQ ID NO: 8, and SEQ ID NO: 17, respectively. In one aspect the heavy chain variable domain comprises the amino acid sequence as set forth in SEQ ID NO: 18. Also disclosed herein are anti-SDCBP binding molecules, wherein the anti-SDCBP binding molecule further comprises a light chain variable domain complementary determining region (CDR) 1 (CDR1), CDR2, and CDR3 as set forth in SEQ ID NO: 11, SEQ ID NO: 12, and SEQ ID NO: 13, respectively (such as, for example an anti-SDCBP binding molecule comprising a light chain variable domain comprising the amino acid sequence as set forth in SEQ ID NO: 10) or SEQ ID NO: 11, SEQ ID NO: 12, and SEQ ID NO: 14, respectively. In one aspect the light chain variable domain comprises the amino acid sequence as set forth in SEQ ID NO: 19.

1. Homology/identity

[0047] It is understood that one way to define any known variants and derivatives or those that might arise, of the

disclosed genes and proteins herein is through defining the variants and derivatives in terms of homology to specific known sequences. For example, SEQ ID NO: 6 sets forth a particular sequence of an anti-SDCBP variable domain heavy chain and SEQ ID NO: 10 sets forth a particular sequence of an anti-SDCBP variable domain light chain. Specifically disclosed are variants of these and other genes and proteins herein disclosed which have at least, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99 percent homology to the stated sequence. Those of skill in the art readily understand how to determine the homology of two proteins or nucleic acids, such as genes. For example, the homology can be calculated after aligning the two sequences so that the homology is at its highest level.

[0048] Another way of calculating homology can be performed by published algorithms. Optimal alignment of sequences for comparison may be conducted by the local homology algorithm of Smith and Waterman *Adv. Appl. Math.* 2: 482 (1981), by the homology alignment algorithm of Needleman and Wunsch, *J. Mol. Biol.* 48: 443 (1970), by the search for similarity method of Pearson and Lipman, *Proc. Natl. Acad. Sci. U.S.A.* 85: 2444 (1988), by computerized implementations of these algorithms (GAP, BEST-FIT, FASTA, and TFASTA in the Wisconsin Genetics Software Package, Genetics Computer Group, 575 Science Dr., Madison, WI), or by inspection.

[0049] The same types of homology can be obtained for nucleic acids by for example the algorithms disclosed in Zuker, M. *Science* 244:48-52, 1989, Jaeger et al. *Proc. Natl. Acad. Sci. USA* 86:7706-7710, 1989, Jaeger et al. *Methods Enzymol.* 183:281-306, 1989 which are herein incorporated by reference for at least material related to nucleic acid alignment.

2. Peptides

A) Protein Variants

[0050] As discussed herein there are numerous variants of the SDCBP protein and anti-SDCBP binding molecules that are known and herein contemplated. Protein variants and derivatives are well understood to those of skill in the art and in can involve amino acid sequence modifications. For example, amino acid sequence modifications typically fall into one or more of three classes: substitutional, insertional or deletional variants. Insertions include amino and/or carboxyl terminal fusions as well as intrasequence insertions of single or multiple amino acid residues. Insertions ordinarily will be smaller insertions than those of amino or carboxyl terminal fusions, for example, on the order of one to four residues. Immunogenic fusion protein derivatives, such as those described in the examples, are made by fusing a polypeptide sufficiently large to confer immunogenicity to the target sequence by cross-linking in vitro or by recombinant cell culture transformed with DNA encoding the fusion. Deletions are characterized by the removal of one or more amino acid residues from the protein sequence. Typically, no more than about from 2 to 6 residues are deleted at any one site within the protein molecule. These variants ordinarily are prepared by site specific mutagenesis of nucleotides in the DNA encoding the protein, thereby producing DNA encoding the variant, and thereafter expressing the DNA in recombinant cell culture. Techniques for making substitution mutations at predetermined sites in DNA having

a known sequence are well known, for example M13 primer mutagenesis and PCR mutagenesis. Amino acid substitutions are typically of single residues, but can occur at a number of different locations at once; insertions usually will be on the order of about from 1 to 10 amino acid residues; and deletions will range about from 1 to 30 residues. Deletions or insertions preferably are made in adjacent pairs, i.e. a deletion of 2 residues or insertion of 2 residues. Substitutions, deletions, insertions or any combination thereof may be combined to arrive at a final construct. The mutations must not place the sequence out of reading frame and preferably will not create complementary regions that could produce secondary mRNA structure. Substitutional variants are those in which at least one residue has been removed and a different residue inserted in its place. Such substitutions generally are made in accordance with the following Tables 1 and 2 and are referred to as conservative substitutions.

TABLE 1

Amino Acid Abbreviations		
Amino Acid	Abbreviations	
Alanine	Ala	A
alloseleucine	Alle	
Arginine	Arg	R
asparagine	Asn	N
aspartic acid	Asp	D
Cysteine	Cys	C
glutamic acid	Glu	E
Glutamine	Gln	Q
Glycine	Gly	G
Histidine	His	H
Isoleucine	Ile	I
Leucine	Leu	L
Lysine	Lys	K
phenylalanine	Phe	F
proline	Pro	P
pyroglutamic acid	pGlu	
Serine	Ser	S
Threonine	Thr	T
Tyrosine	Tyr	Y
Tryptophan	Trp	W
Valine	Val	V

TABLE 2

Amino Acid Substitutions		
Original Residue	Exemplary Conservative Substitutions, others are known in the art.	
Ala	Ser	
Arg	Lys; Gln	
Asn	Gln; His	
Asp	Glu	
Cys	Ser	
Gln	Asn, Lys	
Glu	Asp	
Gly	Pro	
His	Asn; Gln	
Ile	Leu; Val	
Leu	Ile; Val	
Lys	Arg; Gln	
Met	Leu; Ile	
Phe	Met; Leu; Tyr	
Ser	Thr	
Thr	Ser	
Trp	Tyr	
Tyr	Trp; Phe	
Val	Ile; Leu	

[0051] Substantial changes in function or immunological identity are made by selecting substitutions that are less conservative than those in Table 2, i.e., selecting residues that differ more significantly in their effect on maintaining (a) the structure of the polypeptide backbone in the area of the substitution, for example as a sheet or helical conformation, (b) the charge or hydrophobicity of the molecule at the target site or (c) the bulk of the side chain. The substitutions which in general are expected to produce the greatest changes in the protein properties will be those in which (a) a hydrophilic residue, e.g. seryl or threonyl, is substituted for (or by) a hydrophobic residue, e.g. leucyl, isoleucyl, phenylalanyl, valyl or alanyl; (b) a cysteine or proline is substituted for (or by) any other residue; (c) a residue having an electropositive side chain, e.g., lysyl, arginyl, or histidyl, is substituted for (or by) an electronegative residue, e.g., glutamyl or aspartyl; or (d) a residue having a bulky side chain, e.g., phenylalanine, is substituted for (or by) one not having a side chain, e.g., glycine, in this case, (e) by increasing the number of sites for sulfation and/or glycosylation.

[0052] For example, the replacement of one amino acid residue with another that is biologically and/or chemically similar is known to those skilled in the art as a conservative substitution. For example, a conservative substitution would be replacing one hydrophobic residue for another, or one polar residue for another. The substitutions include combinations such as, for example, Gly, Ala; Val, Ile, Leu; Asp, Glu; Asn, Gln; Ser, Thr; Lys, Arg; and Phe, Tyr. Such conservatively substituted variations of each explicitly disclosed sequence are included within the mosaic polypeptides provided herein.

[0053] Substitutional or deletional mutagenesis can be employed to insert sites for N-glycosylation (Asn-X-Thr/Ser) or O-glycosylation (Ser or Thr). Deletions of cysteine or other labile residues also may be desirable. Deletions or substitutions of potential proteolysis sites, e.g. Arg, is accomplished for example by deleting one of the basic residues or substituting one by glutaminyl or histidyl residues.

[0054] Certain post-translational derivatizations are the result of the action of recombinant host cells on the expressed polypeptide. Glutaminyl and asparaginyl residues are frequently post-translationally deamidated to the corresponding glutamyl and aspartyl residues. Alternatively, these residues are deamidated under mildly acidic conditions. Other post-translational modifications include hydroxylation of proline and lysine, phosphorylation of hydroxyl groups of seryl or threonyl residues, methylation of the o-amino groups of lysine, arginine, and histidine side chains (T. E. Creighton, *Proteins: Structure and Molecular Properties*, W. H. Freeman & Co., San Francisco pp 79-86 [1983]), acetylation of the N-terminal amine and, in some instances, amidation of the C-terminal carboxyl.

[0055] It is understood that one way to define the variants and derivatives of the disclosed proteins herein is through defining the variants and derivatives in terms of homology/identity to specific known sequences. For example, SEQ ID NO: 6 sets forth a particular sequence of an anti-SDCBP variable domain heavy chain and SEQ ID NO: 10 sets forth a particular sequence of an anti-SDCBP variable domain light chain. Specifically disclosed are variants of these and other proteins herein disclosed which have at least, 70% or 75% or 80% or 85% or 90% or 95% homology to the stated

sequence. Those of skill in the art readily understand how to determine the homology of two proteins. For example, the homology can be calculated after aligning the two sequences so that the homology is at its highest level.

[0056] Another way of calculating homology can be performed by published algorithms. Optimal alignment of sequences for comparison may be conducted by the local homology algorithm of Smith and Waterman *Adv. Appl. Math.* 2: 482 (1981), by the homology alignment algorithm of Needleman and Wunsch, *J. Mol. Biol.* 48: 443 (1970), by the search for similarity method of Pearson and Lipman, *Proc. Natl. Acad. Sci. U.S.A.* 85: 2444 (1988), by computerized implementations of these algorithms (GAP, BEST-FIT, FASTA, and TFASTA in the Wisconsin Genetics Software Package, Genetics Computer Group, 575 Science Dr., Madison, WI), or by inspection.

[0057] The same types of homology can be obtained for nucleic acids by for example the algorithms disclosed in Zuker, M. *Science* 244:48-52, 1989, Jaeger et al. *Proc. Natl. Acad. Sci. USA* 86:7706-7710, 1989, Jaeger et al. *Methods Enzymol.* 183:281-306, 1989.

[0058] It is understood that the description of conservative mutations and homology can be combined together in any combination, such as embodiments that have at least 70% homology to a particular sequence wherein the variants are conservative mutations.

[0059] As this specification discusses various proteins and protein sequences it is understood that the nucleic acids that can encode those protein sequences are also disclosed. This would include all degenerate sequences related to a specific protein sequence, i.e. all nucleic acids having a sequence that encodes one particular protein sequence as well as all nucleic acids, including degenerate nucleic acids, encoding the disclosed variants and derivatives of the protein sequences. Thus, while each particular nucleic acid sequence may not be written out herein, it is understood that each and every sequence is in fact disclosed and described herein through the disclosed protein sequence. It is also understood that while no amino acid sequence indicates what particular DNA sequence encodes that protein within an organism, where particular variants of a disclosed protein are disclosed herein, the known nucleic acid sequence that encodes that protein is also known and herein disclosed and described.

[0060] It is understood that there are numerous amino acid and peptide analogs which can be incorporated into the disclosed compositions. For example, there are numerous D amino acids or amino acids which have a different functional substituent than the amino acids shown in Table 1 and Table 2. The opposite stereo isomers of naturally occurring peptides are disclosed, as well as the stereo isomers of peptide analogs. These amino acids can readily be incorporated into polypeptide chains by charging tRNA molecules with the amino acid of choice and engineering genetic constructs that utilize, for example, amber codons, to insert the analog amino acid into a peptide chain in a site specific way.

[0061] Molecules can be produced that resemble peptides, but which are not connected via a natural peptide linkage. For example, linkages for amino acids or amino acid analogs can include CH_2NH —, $\text{—CH}_2\text{S—}$, $\text{—CH}_2\text{—CH}_2\text{—}$, —CH=CH— (cis and trans), $\text{—COCH}_2\text{—}$, $\text{—CH(OH)CH}_2\text{—}$, and $\text{—CHH}_2\text{SO—}$ (These and others can be found in Spatola, A. F. in *Chemistry and Biochemistry of Amino Acids, Peptides, and Proteins*, B. Weinstein, eds., Marcel

Dekker, New York, p. 267 (1983); Spatola, A. F., *Vega Data* (March 1983), Vol. 1, Issue 3, Peptide Backbone Modifications (general review); Morley, *Trends Pharm Sci* (1980) pp. 463-468; Hudson, D. et al., *Int J Pept Prot Res* 14:177-185 (1979) ($\text{—CH}_2\text{NH—}$, $\text{CH}_2\text{CH}_2\text{—}$); Spatola et al. *Life Sci* 38:1243-1249 (1986) ($\text{—CHH}_2\text{—S—}$); Hann *J. Chem. Soc Perkin Trans.* 1307-314 (1982) (—CH—CH— , cis and trans); Almquist et al. *J. Med. Chem.* 23:1392-1398 (1980) ($\text{—COCH}_2\text{—}$); Jennings-White et al. *Tetrahedron Lett* 23:2533 (1982) ($\text{—COCH}_2\text{—}$); Szelke et al. *European Appln*, EP 45665 CA (1982); 97:39405 (1982) ($\text{—CH(OH)CH}_2\text{—}$); Holladay et al. *Tetrahedron. Lett* 24:4401-4404 (1983) ($\text{—C(OH)CH}_2\text{—}$); and Hruby *Life Sci* 31:189-199 (1982) ($\text{—CH}_2\text{—S—}$); each of which is incorporated herein by reference. A particularly preferred non-peptide linkage is $\text{—CH}_2\text{NH—}$. It is understood that peptide analogs can have more than one atom between the bond atoms, such as b-alanine, g-aminobutyric acid, and the like.

[0062] Amino acid analogs and analogs and peptide analogs often have enhanced or desirable properties, such as, more economical production, greater chemical stability, enhanced pharmacological properties (half-life, absorption, potency, efficacy, etc.), altered specificity (e.g., a broad-spectrum of biological activities), reduced antigenicity, and others.

[0063] D-amino acids can be used to generate more stable peptides, because D amino acids are not recognized by peptidases and such. Systematic substitution of one or more amino acids of a consensus sequence with a D-amino acid of the same type (e.g., D-lysine in place of L-lysine) can be used to generate more stable peptides. Cysteine residues can be used to cyclize or attach two or more peptides together. This can be beneficial to constrain peptides into particular conformations.

3. Immunoassays and Fluorochromes

[0064] The steps of various useful immunodetection methods have been described in the scientific literature, such as, e.g., Maggio et al., *Enzyme-Immunoassay*, (1987) and Nakamura, et al., *Enzyme Immunoassays: Heterogeneous and Homogeneous Systems, Handbook of Experimental Immunology*, Vol. 1: Immunochemistry, 27.1-27.20 (1986), each of which is incorporated herein by reference in its entirety and specifically for its teaching regarding immunodetection methods. Immunoassays, in their most simple and direct sense, are binding assays involving binding between antibodies and antigen. Many types and formats of immunoassays are known and all are suitable for detecting the disclosed biomarkers. Examples of immunoassays are enzyme linked immunosorbent assays (ELISAs), radioimmunoassays (RIA), radioimmune precipitation assays (RIPA), immunobead capture assays, Western blotting, dot blotting, gel-shift assays, Flow cytometry, protein arrays, multiplexed bead arrays, magnetic capture, in vivo imaging, fluorescence resonance energy transfer (FRET), and fluorescence recovery/localization after photobleaching (FRAP/FLAP).

[0065] In general, immunoassays involve contacting a sample suspected of containing a molecule of interest (such as the disclosed biomarkers) with an antibody to the molecule of interest or contacting an antibody to a molecule of interest (such as antibodies to the disclosed biomarkers) with a molecule that can be bound by the antibody, as the case may be, under conditions effective to allow the formation of

immunocomplexes. Contacting a sample with the antibody to the molecule of interest or with the molecule that can be bound by an antibody to the molecule of interest under conditions effective and for a period of time sufficient to allow the formation of immune complexes (primary immune complexes) is generally a matter of simply bringing into contact the molecule or antibody and the sample and incubating the mixture for a period of time long enough for the antibodies to form immune complexes with, i.e., to bind to, any molecules (e.g., antigens) present to which the antibodies can bind. In many forms of immunoassay, the sample-antibody composition, such as a tissue section, ELISA plate, dot blot or Western blot, can then be washed to remove any non-specifically bound antibody species, allowing only those antibodies specifically bound within the primary immune complexes to be detected.

[0066] Immunoassays can include methods for detecting or quantifying the amount of a molecule of interest (such as the disclosed biomarkers or their antibodies) in a sample, which methods generally involve the detection or quantitation of any immune complexes formed during the binding process. In general, the detection of immunocomplex formation is well known in the art and can be achieved through the application of numerous approaches. These methods are generally based upon the detection of a label or marker, such as any radioactive, fluorescent, biological or enzymatic tags or any other known label.

[0067] As used herein, a label can include a fluorescent dye, a member of a binding pair, such as biotin/streptavidin, a metal (e.g., gold), or an epitope tag that can specifically interact with a molecule that can be detected, such as by producing a colored substrate or fluorescence. Substances suitable for detectably labeling proteins include fluorescent dyes (also known herein as fluorochromes and fluorophores) and enzymes that react with colorimetric substrates (e.g., horseradish peroxidase). The use of fluorescent dyes is generally preferred in the practice of the invention as they can be detected at very low amounts. Furthermore, in the case where multiple antigens are reacted with a single array, each antigen can be labeled with a distinct fluorescent compound for simultaneous detection. Labeled spots on the array are detected using a fluorimeter, the presence of a signal indicating an antigen bound to a specific antibody.

[0068] Fluorophores are compounds or molecules that luminesce. Typically fluorophores absorb electromagnetic energy at one wavelength and emit electromagnetic energy at a second wavelength. Representative fluorophores include, but are not limited to, 1,5 IAEDANS; 1,8-ANS; 4-Methylumbelliferone; 5-carboxy-2,7-dichlorofluorescein; 5-Carboxyfluorescein (5-FAM); 5-Carboxynaphthofluorescein; 5-Carboxytetramethylrhodamine (5-TAMRA); 5-Hydroxy Tryptamine (5-HAT); 5-ROX (carboxy-X-rhodamine); 6-Carboxyrhodamine 6G; 6-CR 6G; 6-JOE; 7-Amino-4-methylcoumarin; 7-Aminoactinomycin D (7-AAD); 7-Hydroxy-4-I methylcoumarin; 9-Amino-6-chloro-2-methoxyacridine (ACMA); ABQ; Acid Fuchsin; Acridine Orange; Acridine Red; Acridine Yellow; Acriflavin; Acriflavin Feulgen SITSA; Aequorin (Photoprotein); AFPs—AutoFluorescent Protein—(Quantum Biotechnologies) see sgGFP, sgBFP; Alexa Fluor 350™; Alexa Fluor 430™; Alexa Fluor 488™; Alexa Fluor 532™; Alexa Fluor 546™; Alexa Fluor 568™; Alexa Fluor 594™; Alexa Fluor 633™; Alexa Fluor 647™; Alexa Fluor 660™; Alexa Fluor 680™; Alizarin Complexon; Alizarin Red; Allophycocyanin

(APC); AMC, AMCA-S; Aminomethylcoumarin (AMCA); AMCA-X; Aminoactinomycin D; Aminocoumarin; Anilin Blue; Anthrocyll stearate; APC-Cy7; APTRA-BTC; APTS; Astrazon Brilliant Red 4G; Astrazon Orange R; Astrazon Red 6B; Astrazon Yellow 7 GLL; Atabrine; ATTO-TAG™ CBQCA; ATTO-TAG™ FQ; Auramine; Auorophosphine G; Auorophosphine; BAO 9 (Bisaininophenylloxadiazole); BCECF (high pH); BCECF (low pH); Berberine Sulphate; Beta Lactamase; BFP blue shifted GFP (Y66H); Blue Fluorescent Protein; BFP/GFP FRET; Bimane; Bisbenzamide; Bisbenzimidazole (Hoechst); bis-BTC; Blancophor FFG; Blancophor SV; BOBO™-1; BOBO™-3; Bodipy492/515; Bodipy493/503; Bodipy500/510; Bodipy; 505/515; Bodipy 530/550; Bodipy 542/563; Bodipy 558/568; Bodipy 564/570; Bodipy 576/589; Bodipy 581/591; Bodipy 630/650-X; Bodipy 650/665-X; Bodipy 665/676; Bodipy FI; Bodipy FL ATP; Bodipy F1-Ceramide; Bodipy R6G SE; Bodipy TMR; Bodipy TMR-X conjugate; Bodipy TMR-X, SE; Bodipy TR; Bodipy TR ATP; Bodipy TR-X SE; BO-PRO™-1; BO-PRO™-3; Brilliant Sulphoflavin FF; BTC; BTC-5N; Calcein; Calcein Blue; Calcium Crimson-; Calcium Green; Calcium Green-1 Ca²⁺ Dye; Calcium Green-2 Ca²⁺; Calcium Green-5N Ca²⁺; Calcium Green-C18 Ca²⁺; Calcium Orange; Calcofluor White; Carboxy-X-rhodamine (5-ROX); Cascade Blue™; Cascade Yellow; Catecholamine; CCF2 (GeneBlazer); CFDA; CFP (Cyan Fluorescent Protein); CFP/YFP FRET; Chlorophyll; Chromomycin A; Chromomycin A; CL-NERF; CMFDA; Coelenterazine; Coelenterazine cp; Coelenterazine f; Coelenterazine fcp; Coelenterazine h; Coelenterazine hcp; Coelenterazine ip; Coelenterazine n; Coelenterazine O; Coumarin Phalloidin; C-phycocyanine; CPM I Methylcoumarin; CTC; CTC Formazan; Cy2™; Cy3.1 8; Cy3.5™; Cy3™; Cy5.1 8; Cy5.5™; Cy5™; Cy7™; Cyan GFP; cyclic AMP Fluorosensor (FicRHR); Dabcyl; Dansyl; Dansyl Amine; Dansyl Cadaverine; Dansyl Chloride; Dansyl DHPE; Dansyl fluoride; DAPI; Dapoxyl; Dapoxyl 2; Dapoxyl 3'DCFDA; DCFH (Dichlorodihydrofluorescein Diacetate); DDAO; DHR (Dihydrodihydrofluorescein 123); Di-4-ANEPPS; Di-8-ANEPPS (non-ratio); DiA (4-Di 16-ASP); Dichlorodihydrofluorescein Diacetate (DCFH); DiD-Lipophilic Tracer; DiD (DiI18(5)); DIDS; Dihydrodihydrofluorescein 123 (DHR); DiI (DiI18(3)); I Dinitrophenol; DiO (DiO18(3)); DiR; DiR (DiI18(7)); DM-NERF (high pH); DNP; Dopamine; DsRed; DTAF; DY-630-NHS; DY-635-NHS; EBFP; ECFP; EGFP; ELF 97; Eosin; Erythrosin; Erythrosin ITC; Ethidium Bromide; Ethidium homodimer-1 (EthD-1); Euchrysin; EukoLight; Europium (111) chloride; EYFP; Fast Blue; FDA; Feulgen (Pararosaniline); FIF (Formaldehyde Induced Fluorescence); FITC; Flazo Orange; Fluo-3; Fluo-4; Fluorescein (FITC); Fluorescein Diacetate; Fluoro-Emerald; Fluoro-Gold (Hydroxystilbamidine); Fluor-Ruby; FluorX; FM 1-43™; FM 4-46; Fura Red™ (high pH); Fura Red™/Fluo-3; Fura-2; Fura-2/BCECF; Genacryl Brilliant Red B; Genacryl Brilliant Yellow 10GF; Genacryl Pink 3G; Genacryl Yellow 5GF; GeneBlazer; (CCF2); GFP (S65T); GFP red shifted (rsGFP); GFP wild type* non-UV excitation (wtGFP); GFP wild type, UV excitation (wtGFP); GFPuv; Gloxalic Acid; Granular blue; Haematoporphyrin; Hoechst 33258; Hoechst 33342; Hoechst 34580; HPTS; Hydroxycoumarin; Hydroxystilbamidine (FluoroGold); Hydroxytryptamine; Indo-1, high calcium; Indo-1 low calcium; Indodicarbocyanine (DiD); Indotricarbocyanine (DiR); Intrawhite Cf; JC-1; JO JO-1; JO-PRO-1; LaserPro;

Laurodan; LDS 751 (DNA); LDS 751 (RNA); Leucophor PAF; Leucophor SF; Leucophor WS; Lissamine Rhodamine; Lissamine Rhodamine B; Calcein/Ethidium homodimer; LOLO-1; LO-PRO-1; Lucifer Yellow; Lyso Tracker Blue; Lyso Tracker Blue-White; Lyso Tracker Green; Lyso Tracker Red; Lyso Tracker Yellow; LysoSensor Blue; LysoSensor Green; LysoSensor Yellow/Blue; Mag Green; Magdala Red (Phloxin B); Mag-Fura Red; Mag-Fura-2; Mag-Fura-5; Mag-Indo-1; Magnesium Green; Magnesium Orange; Malachite Green; Marina Blue; I Maxilon Brilliant Flavin 10 GFF; Maxilon Brilliant Flavin 8 GFF; Merocyanin; Methoxycoumarin; Mitotracker Green FM; Mitotracker Orange; Mitotracker Red; Mitracycline; Monobromobimane; Monobromobimane (mBBBr-GSH); Monochlorobimane; MPS (Methyl Green Pyronine Stilbene); NBD; NBD Amine; Nile Red; Nitrobenzoxedidole; Noradrenaline; Nuclear Fast Red; i Nuclear Yellow; Nylosan Brilliant lavin E8G; Oregon Green™; Oregon Green™ 488; Oregon Green™ 500; Oregon Green™ 514; Pacific Blue; Pararosaniline (Feulgen); PBFI; PE-Cy5; PE-Cy7; PerCP; PerCP-Cy5.5; PE-TexasRed (Red 613); Phloxin B (Magdala Red); Phorwite AR; Phorwite BKL; Phorwite Rev; Phorwite RPA; Phosphine 3R; PhotoResist; Phycoerythrin B [PE]; Phycoerythrin R [PE]; PKH26 (Sigma); PKH67; PMIA; Pontochrome Blue Black; POPO-1; POPO-3; PO-PRO-1; PO-I PRO-3; Primuline; Procion Yellow; Propidium Iodid (P1); PyMPO; Pyrene; Pyronine; Pyronine B; Pyroزال Brilliant Flavin 7GF; QSY 7; Quinacrine Mustard; Resorufin; RH 414; Rhod-2; Rhodamine; Rhodamine 110; Rhodamine 123; Rhodamine 5 GLD; Rhodamine 6G; Rhodamine B; Rhodamine B 200; Rhodamine B extra; Rhodamine BB; Rhodamine BG; Rhodamine Green; Rhodamine Phalloidine; Rhodamine: Phalloidine; Rhodamine Red; Rhodamine WT; Rose Bengal; R-phycoerythrin; R-phycoerythrin (PE); rsGFP; S65A; S65C; S65L; S65T; Sapphire GFP; SBFI; Serotonin; Sevron Brilliant Red 2B; Sevron Brilliant Red 4G; Sevron I Brilliant Red B; Sevron Orange; Sevron Yellow L; sgBFP™ (super glow BFP); sgGFP™ (super glow GFP); SITS (Primuline; Stilbene Isothiosulphonic Acid); SNAFL calcein; SNAFL-1; SNAFL-2; SNARF calcein; SNARF; Sodium Green; SpectrumAqua; SpectrumGreen; SpectrumOrange; Spectrum Red; SPQ (6-methoxy-N-(3 sulfopropyl) quinolinium); Stilbene; Sulphorhodamine B and C; Sulphorhodamine Extra; SYTO 11; SYTO 12; SYTO 13; SYTO 14; SYTO 15; SYTO 16; SYTO 17; SYTO 18; SYTO 20; SYTO 21; SYTO 22; SYTO 23; SYTO 24; SYTO 25; SYTO 40; SYTO 41; SYTO 42; SYTO 43; SYTO 44; SYTO 45; SYTO 59; SYTO 60; SYTO 61; SYTO 62; SYTO 63; SYTO 64; SYTO 80; SYTO 81; SYTO 82; SYTO 83; SYTO 84; SYTO 85; SYTOX Blue; SYTOX Green; SYTOX Orange; Tetracycline; Tetramethylrhodamine (TRITC); Texas Red™; Texas Red-X™ conjugate; Thiadiazocyanine (DiSC3); Thiazine Red R; Thiazole Orange; Thioflavin 5; Thioflavin S; Thioflavin TON; Thioylte; Thiozole Orange; Tinopol CBS (Calcofluor White); TIER; TO-PRO-1; TO-PRO-3; TO-PRO-5; TOTO-1; TOTO-3; TriColor (PE-Cy5); TRITC TetramethylRhodamine; TrueBlue; True Red; Ultralite; Uranine B; Uvitex SFC; wt GFP; WW 781; X-Rhodamine; XRITC; Xylene Orange; Y66F; Y66H; Y66W; Yellow GFP; YFP; YO-PRO-1; YO-PRO 3; YOYO-1; YOYO-3; Sybr Green; Thiazole orange (interchelating dyes); semiconductor nanoparticles such as quantum dots; or caged fluorophore

(which can be activated with light or other electromagnetic energy source), or a combination thereof.

[0069] A modifier unit such as a radionuclide can be incorporated into or attached directly to any of the compounds described herein by halogenation. Examples of radionuclides useful in this embodiment include, but are not limited to, tritium, iodine-125, iodine-131, iodine-123, iodine-124, astatine-210, carbon-11, carbon-14, nitrogen-13, fluorine-18. In another aspect, the radionuclide can be attached to a linking group or bound by a chelating group, which is then attached to the compound directly or by means of a linker. Examples of radionuclides useful in the aspect include, but are not limited to, Tc-99m, Re-186, Ga-68, Re-188, Y-90, Sm-153, Bi-212, Cu-67, Cu-64, and Cu-62. Radiolabeling techniques such as these are routinely used in the radiopharmaceutical industry.

[0070] The radiolabeled compounds are useful as imaging agents to diagnose neurological disease (e.g., a neurodegenerative disease) or a mental condition or to follow the progression or treatment of such a disease or condition in a mammal (e.g., a human). The radiolabeled compounds described herein can be conveniently used in conjunction with imaging techniques such as positron emission tomography (PET) or single photon emission computerized tomography (SPECT).

[0071] Labeling can be either direct or indirect. In direct labeling, the detecting antibody (the antibody for the molecule of interest) or detecting molecule (the molecule that can be bound by an antibody to the molecule of interest) include a label. Detection of the label indicates the presence of the detecting antibody or detecting molecule, which in turn indicates the presence of the molecule of interest or of an antibody to the molecule of interest, respectively. In indirect labeling, an additional molecule or moiety is brought into contact with, or generated at the site of, the immunocomplex. For example, a signal-generating molecule or moiety such as an enzyme can be attached to or associated with the detecting antibody or detecting molecule. The signal-generating molecule can then generate a detectable signal at the site of the immunocomplex. For example, an enzyme, when supplied with suitable substrate, can produce a visible or detectable product at the site of the immunocomplex. ELISAs use this type of indirect labeling.

[0072] As another example of indirect labeling, an additional molecule (which can be referred to as a binding agent) that can bind to either the molecule of interest or to the antibody (primary antibody) to the molecule of interest, such as a second antibody to the primary antibody, can be contacted with the immunocomplex. The additional molecule can have a label or signal-generating molecule or moiety. The additional molecule can be an antibody, which can thus be termed a secondary antibody. Binding of a secondary antibody to the primary antibody can form a so-called sandwich with the first (or primary) antibody and the molecule of interest. The immune complexes can be contacted with the labeled, secondary antibody under conditions effective and for a period of time sufficient to allow the formation of secondary immune complexes. The secondary immune complexes can then be generally washed to remove any non-specifically bound labeled secondary antibodies, and the remaining label in the secondary immune complexes can then be detected. The additional molecule can also be or include one of a pair of molecules or moieties that can bind to each other, such as the biotin/avidin pair. In

this mode, the detecting antibody or detecting molecule should include the other member of the pair.

[0073] Other modes of indirect labeling include the detection of primary immune complexes by a two step approach. For example, a molecule (which can be referred to as a first binding agent), such as an antibody, that has binding affinity for the molecule of interest or corresponding antibody can be used to form secondary immune complexes, as described above. After washing, the secondary immune complexes can be contacted with another molecule (which can be referred to as a second binding agent) that has binding affinity for the first binding agent, again under conditions effective and for a period of time sufficient to allow the formation of immune complexes (thus forming tertiary immune complexes). The second binding agent can be linked to a detectable label or signal-generating molecule or moiety, allowing detection of the tertiary immune complexes thus formed. This system can provide for signal amplification.

[0074] Immunoassays that involve the detection of a substance, such as a protein or an antibody to a specific protein, include label-free assays, protein separation methods (i.e., electrophoresis), solid support capture assays, or in vivo detection. Label-free assays are generally diagnostic means of determining the presence or absence of a specific protein, or an antibody to a specific protein, in a sample. Protein separation methods are additionally useful for evaluating physical properties of the protein, such as size or net charge. Capture assays are generally more useful for quantitatively evaluating the concentration of a specific protein, or antibody to a specific protein, in a sample. Finally, in vivo detection is useful for evaluating the spatial expression patterns of the substance, i.e., where the substance can be found in a subject, tissue or cell.

[0075] Provided that the concentrations are sufficient, the molecular complexes ([Ab-Ag]_n) generated by antibody-antigen interaction are visible to the naked eye, but smaller amounts may also be detected and measured due to their ability to scatter a beam of light. The formation of complexes indicates that both reactants are present, and in immunoprecipitation assays a constant concentration of a reagent antibody is used to measure specific antigen ([Ab-Ag]_n), and reagent antigens are used to detect specific antibody ([Ab-Ag]_n). If the reagent species is previously coated onto cells (as in hemagglutination assay) or very small particles (as in latex agglutination assay), "clumping" of the coated particles is visible at much lower concentrations. A variety of assays based on these elementary principles are in common use, including Ouchterlony immunodiffusion assay, rocket immunoelectrophoresis, and immunoturbidometric and nephelometric assays. The main limitations of such assays are restricted sensitivity (lower detection limits) in comparison to assays employing labels and, in some cases, the fact that very high concentrations of analyte can actually inhibit complex formation, necessitating safeguards that make the procedures more complex. Some of these Group 1 assays date right back to the discovery of antibodies and none of them have an actual "label" (e.g. Ag-enz). Other kinds of immunoassays that are label free depend on immunosensors, and a variety of instruments that can directly detect antibody-antigen interactions are now commercially available. Most depend on generating an evanescent wave on a sensor surface with immobilized ligand, which allows continuous monitoring of binding to the ligand. Immunosensors allow the easy inves-

tigation of kinetic interactions and, with the advent of lower-cost specialized instruments, may in the future find wide application in immunoanalysis.

[0076] The use of immunoassays to detect a specific protein can involve the separation of the proteins by electrophoresis. Electrophoresis is the migration of charged molecules in solution in response to an electric field. Their rate of migration depends on the strength of the field; on the net charge, size and shape of the molecules and also on the ionic strength, viscosity and temperature of the medium in which the molecules are moving. As an analytical tool, electrophoresis is simple, rapid and highly sensitive. It is used analytically to study the properties of a single charged species, and as a separation technique.

[0077] Generally the sample is run in a support matrix such as paper, cellulose acetate, starch gel, agarose or polyacrylamide gel. The matrix inhibits convective mixing caused by heating and provides a record of the electrophoretic run: at the end of the run, the matrix can be stained and used for scanning, autoradiography or storage. In addition, the most commonly used support matrices—agarose and polyacrylamide—provide a means of separating molecules by size, in that they are porous gels. A porous gel may act as a sieve by retarding, or in some cases completely obstructing, the movement of large macromolecules while allowing smaller molecules to migrate freely. Because dilute agarose gels are generally more rigid and easy to handle than polyacrylamide of the same concentration, agarose is used to separate larger macromolecules such as nucleic acids, large proteins and protein complexes. Polyacrylamide, which is easy to handle and to make at higher concentrations, is used to separate most proteins and small oligonucleotides that require a small gel pore size for retardation.

[0078] Proteins are amphoteric compounds; their net charge therefore is determined by the pH of the medium in which they are suspended. In a solution with a pH above its isoelectric point, a protein has a net negative charge and migrates towards the anode in an electrical field. Below its isoelectric point, the protein is positively charged and migrates towards the cathode. The net charge carried by a protein is in addition independent of its size—i.e., the charge carried per unit mass (or length, given proteins and nucleic acids are linear macromolecules) of molecule differs from protein to protein. At a given pH therefore, and under non-denaturing conditions, the electrophoretic separation of proteins is determined by both size and charge of the molecules.

[0079] Sodium dodecyl sulphate (SDS) is an anionic detergent which denatures proteins by "wrapping around" the polypeptide backbone—and SDS binds to proteins fairly specifically in a mass ratio of 1.4:1. In so doing, SDS confers a negative charge to the polypeptide in proportion to its length. Further, it is usually necessary to reduce disulphide bridges in proteins (denature) before they adopt the random-coil configuration necessary for separation by size; this is done with 2-mercaptoethanol or dithiothreitol (DTT). In denaturing SDS-PAGE separations therefore, migration is determined not by intrinsic electrical charge of the polypeptide, but by molecular weight.

[0080] Determination of molecular weight is done by SDS-PAGE of proteins of known molecular weight along with the protein to be characterized. A linear relationship exists between the logarithm of the molecular weight of an SDS-denatured polypeptide, or native nucleic acid, and its

Rf. The Rf is calculated as the ratio of the distance migrated by the molecule to that migrated by a marker dye-front. A simple way of determining relative molecular weight by electrophoresis (Mr) is to plot a standard curve of distance migrated vs. log 10 MW for known samples, and read off the log Mr of the sample after measuring distance migrated on the same gel.

[0081] In two-dimensional electrophoresis, proteins are fractionated first on the basis of one physical property, and, in a second step, on the basis of another. For example, isoelectric focusing can be used for the first dimension, conveniently carried out in a tube gel, and SDS electrophoresis in a slab gel can be used for the second dimension. One example of a procedure is that of O'Farrell, P. H., High Resolution Two-dimensional Electrophoresis of Proteins, *J. Biol. Chem.* 250:4007-4021 (1975), herein incorporated by reference in its entirety for its teaching regarding two-dimensional electrophoresis methods. Other examples include but are not limited to, those found in Anderson, L and Anderson, NG, High resolution two-dimensional electrophoresis of human plasma proteins, *Proc. Natl. Acad. Sci.* 74:5421-5425 (1977), Ornstein, L., Disc electrophoresis, *L. Ann. N.Y. Acad. Sci.* 121:321349 (1964), each of which is herein incorporated by reference in its entirety for teachings regarding electrophoresis methods. Laemmli, U.K., Cleavage of structural proteins during the assembly of the head of bacteriophage T4, *Nature* 227:680 (1970), which is herein incorporated by reference in its entirety for teachings regarding electrophoresis methods, discloses a discontinuous system for resolving proteins denatured with SDS. The leading ion in the Laemmli buffer system is chloride, and the trailing ion is glycine. Accordingly, the resolving gel and the stacking gel are made up in Tris-HCl buffers (of different concentration and pH), while the tank buffer is Tris-glycine. All buffers contain 0.1% SDS.

[0082] One example of an immunoassay that uses electrophoresis that is contemplated in the current methods is Western blot analysis. Western blotting or immunoblotting allows the determination of the molecular mass of a protein and the measurement of relative amounts of the protein present in different samples. Detection methods include chemiluminescence and chromagenic detection. Standard methods for Western blot analysis can be found in, for example, D. M. Bollag et al., *Protein Methods* (2d edition 1996) and E. Harlow & D. Lane, *Antibodies, a Laboratory Manual* (1988), U.S. Pat. No. 4,452,901, each of which is herein incorporated by reference in its entirety for teachings regarding Western blot methods. Generally, proteins are separated by gel electrophoresis, usually SDS-PAGE. The proteins are transferred to a sheet of special blotting paper, e.g., nitrocellulose, though other types of paper, or membranes, can be used. The proteins retain the same pattern of separation they had on the gel. The blot is incubated with a generic protein (such as milk proteins) to bind to any remaining sticky places on the nitrocellulose. An antibody is then added to the solution which is able to bind to its specific protein.

[0083] The attachment of specific antibodies to specific immobilized antigens can be readily visualized by indirect enzyme immunoassay techniques, usually using a chromogenic substrate (e.g. alkaline phosphatase or horseradish peroxidase) or chemiluminescent substrates. Other possibilities for probing include the use of fluorescent or radioisotope labels (e.g., fluorescein, ¹²⁵I). Probes for the detection of

antibody binding can be conjugated anti-immunoglobulins, conjugated staphylococcal Protein A (binds IgG), or probes to biotinylated primary antibodies (e.g., conjugated avidin/streptavidin).

[0084] The power of the technique lies in the simultaneous detection of a specific protein by means of its antigenicity, and its molecular mass. Proteins are first separated by mass in the SDS-PAGE, then specifically detected in the immunoassay step. Thus, protein standards (ladders) can be run simultaneously in order to approximate molecular mass of the protein of interest in a heterogeneous sample.

[0085] The gel shift assay or electrophoretic mobility shift assay (EMSA) can be used to detect the interactions between DNA binding proteins and their cognate DNA recognition sequences, in both a qualitative and quantitative manner. Exemplary techniques are described in Ornstein L., Disc electrophoresis-I: Background and theory, *Ann. NY Acad. Sci.* 121:321-349 (1964), and Matsudaira, PT and DR Burgess, SDS microslab linear gradient polyacrylamide gel electrophoresis, *Anal. Biochem.* 87:386-396 (1987), each of which is herein incorporated by reference in its entirety for teachings regarding gel-shift assays.

[0086] In a general gel-shift assay, purified proteins or crude cell extracts can be incubated with a labeled (e.g., ³²P-radiolabeled) DNA or RNA probe, followed by separation of the complexes from the free probe through a non-denaturing polyacrylamide gel. The complexes migrate more slowly through the gel than unbound probe. Depending on the activity of the binding protein, a labeled probe can be either double-stranded or single-stranded. For the detection of DNA binding proteins such as transcription factors, either purified or partially purified proteins, or nuclear cell extracts can be used. For detection of RNA binding proteins, either purified or partially purified proteins, or nuclear or cytoplasmic cell extracts can be used. The specificity of the DNA or RNA binding protein for the putative binding site is established by competition experiments using DNA or RNA fragments or oligonucleotides containing a binding site for the protein of interest, or other unrelated sequence. The differences in the nature and intensity of the complex formed in the presence of specific and nonspecific competitor allows identification of specific interactions. Refer to Promega, Gel Shift Assay FAQ, available at <<http://www.promega.com/faq/gelshfaq.html>> (last visited Mar. 25, 2005), which is herein incorporated by reference in its entirety for teachings regarding gel shift methods.

[0087] Gel shift methods can include using, for example, colloidal forms of COOMASSIE (Imperial Chemicals Industries, Ltd) blue stain to detect proteins in gels such as polyacrylamide electrophoresis gels. Such methods are described, for example, in Neuhoff et al., *Electrophoresis* 6:427-448 (1985), and Neuhoff et al., *Electrophoresis* 9:255-262 (1988), each of which is herein incorporated by reference in its entirety for teachings regarding gel shift methods. In addition to the conventional protein assay methods referenced above, a combination cleaning and protein staining composition is described in U.S. Pat. No. 5,424,000, herein incorporated by reference in its entirety for its teaching regarding gel shift methods. The solutions can include phosphoric, sulfuric, and nitric acids, and Acid Violet dye.

[0088] Radioimmune Precipitation Assay (RIPA) is a sensitive assay using radiolabeled antigens to detect specific antibodies in serum. The antigens are allowed to react with the serum and then precipitated using a special reagent such

as, for example, protein A sepharose beads. The bound radiolabeled immunoprecipitate is then commonly analyzed by gel electrophoresis. Radioimmunoprecipitation assay (RIPA) is often used as a confirmatory test for diagnosing the presence of HIV antibodies. RIPA is also referred to in the art as Farr Assay, Precipitin Assay, Radioimmune Precipitin Assay; Radioimmunoprecipitation Analysis; Radioimmunoprecipitation Analysis, and Radioimmunoprecipitation Analysis.

[0089] While the above immunoassays that utilize electrophoresis to separate and detect the specific proteins of interest allow for evaluation of protein size, they are not very sensitive for evaluating protein concentration. However, also contemplated are immunoassays wherein the protein or antibody specific for the protein is bound to a solid support (e.g., tube, well, bead, or cell) to capture the antibody or protein of interest, respectively, from a sample, combined with a method of detecting the protein or antibody specific for the protein on the support. Examples of such immunoassays include Radioimmunoassay (RIA), Enzyme-Linked Immunosorbent Assay (ELISA), Flow cytometry, protein array, multiplexed bead assay, and magnetic capture.

[0090] Radioimmunoassay (RIA) is a classic quantitative assay for detection of antigen-antibody reactions using a radioactively labeled substance (radioligand), either directly or indirectly, to measure the binding of the unlabeled substance to a specific antibody or other receptor system. Radioimmunoassay is used, for example, to test hormone levels in the blood without the need to use a bioassay. Non-immunogenic substances (e.g., haptens) can also be measured if coupled to larger carrier proteins (e.g., bovine gamma-globulin or human serum albumin) capable of inducing antibody formation. RIA involves mixing a radioactive antigen (because of the ease with which iodine atoms can be introduced into tyrosine residues in a protein, the radioactive isotopes ^{125}I or ^{131}I are often used) with antibody to that antigen. The antibody is generally linked to a solid support, such as a tube or beads. Unlabeled or “cold” antigen is then added in known quantities and measuring the amount of labeled antigen displaced. Initially, the radioactive antigen is bound to the antibodies. When cold antigen is added, the two compete for antibody binding sites—and at higher concentrations of cold antigen, more binds to the antibody, displacing the radioactive variant. The bound antigens are separated from the unbound ones in solution and the radioactivity of each used to plot a binding curve. The technique is both extremely sensitive, and specific.

[0091] Enzyme-Linked Immunosorbent Assay (ELISA), or more generically termed EIA (Enzyme ImmunoAssay), is an immunoassay that can detect an antibody specific for a protein. In such an assay, a detectable label bound to either an antibody-binding or antigen-binding reagent is an enzyme. When exposed to its substrate, this enzyme reacts in such a manner as to produce a chemical moiety which can be detected, for example, by spectrophotometric, fluorometric or visual means. Enzymes which can be used to detectably label reagents useful for detection include, but are not limited to, horseradish peroxidase, alkaline phosphatase, glucose oxidase, β -galactosidase, ribonuclease, urease, catalase, malate dehydrogenase, staphylococcal nuclease, asparaginase, yeast alcohol dehydrogenase, alpha-glycerophosphate dehydrogenase, triose phosphate isomerase, glucose-6-phosphate dehydrogenase, glucoamylase and acetylcholinesterase.

[0092] Variations of ELISA techniques are known to those of skill in the art. In one variation, antibodies that can bind to proteins can be immobilized onto a selected surface exhibiting protein affinity, such as a well in a polystyrene microtiter plate. Then, a test composition suspected of containing a marker antigen can be added to the wells. After binding and washing to remove non-specifically bound immunocomplexes, the bound antigen can be detected. Detection can be achieved by the addition of a second antibody specific for the target protein, which is linked to a detectable label. This type of ELISA is a simple “sandwich ELISA.” Detection also can be achieved by the addition of a second antibody, followed by the addition of a third antibody that has binding affinity for the second antibody, with the third antibody being linked to a detectable label.

[0093] Another variation is a competition ELISA. In competition ELISA's, test samples compete for binding with known amounts of labeled antigens or antibodies. The amount of reactive species in the sample can be determined by mixing the sample with the known labeled species before or during incubation with coated wells. The presence of reactive species in the sample acts to reduce the amount of labeled species available for binding to the well and thus reduces the ultimate signal.

[0094] Regardless of the format employed, ELISAs have certain features in common, such as coating, incubating or binding, washing to remove non-specifically bound species, and detecting the bound immunocomplexes. Antigen or antibodies can be linked to a solid support, such as in the form of plate, beads, dipstick, membrane or column matrix, and the sample to be analyzed applied to the immobilized antigen or antibody. In coating a plate with either antigen or antibody, one will generally incubate the wells of the plate with a solution of the antigen or antibody, either overnight or for a specified period of hours. The wells of the plate can then be washed to remove incompletely adsorbed material. Any remaining available surfaces of the wells can then be “coated” with a nonspecific protein that is antigenically neutral with regard to the test antisera. These include bovine serum albumin (BSA), casein and solutions of milk powder. The coating allows for blocking of nonspecific adsorption sites on the immobilizing surface and thus reduces the background caused by nonspecific binding of antisera onto the surface.

[0095] In ELISAs, a secondary or tertiary detection means rather than a direct procedure can also be used. Thus, after binding of a protein or antibody to the well, coating with a non-reactive material to reduce background, and washing to remove unbound material, the immobilizing surface is contacted with the control clinical or biological sample to be tested under conditions effective to allow immunocomplex (antigen/antibody) formation. Detection of the immunocomplex then requires a labeled secondary binding agent or a secondary binding agent in conjunction with a labeled third binding agent.

[0096] Enzyme-Linked Immunospot Assay (ELISPOT) is an immunoassay that can detect an antibody specific for a protein or antigen. In such an assay, a detectable label bound to either an antibody-binding or antigen-binding reagent is an enzyme. When exposed to its substrate, this enzyme reacts in such a manner as to produce a chemical moiety which can be detected, for example, by spectrophotometric, fluorometric or visual means. Enzymes which can be used to detectably label reagents useful for detection include, but are

not limited to, horseradish peroxidase, alkaline phosphatase, glucose oxidase, β -galactosidase, ribonuclease, urease, catalase, malate dehydrogenase, staphylococcal nuclease, asparaginase, yeast alcohol dehydrogenase, α -glycerophosphate dehydrogenase, triose phosphate isomerase, glucose-6-phosphate dehydrogenase, glucoamylase and acetylcholinesterase. In this assay a nitrocellulose microtiter plate is coated with antigen. The test sample is exposed to the antigen and then reacted similarly to an ELISA assay. Detection differs from a traditional ELISA in that detection is determined by the enumeration of spots on the nitrocellulose plate. The presence of a spot indicates that the sample reacted to the antigen. The spots can be counted and the number of cells in the sample specific for the antigen determined.

[0097] “Under conditions effective to allow immunocomplex (antigen/antibody) formation” means that the conditions include diluting the antigens and antibodies with solutions such as BSA, bovine gamma globulin (BGG) and phosphate buffered saline (PBS)/Tween so as to reduce non-specific binding and to promote a reasonable signal to noise ratio.

[0098] The suitable conditions also mean that the incubation is at a temperature and for a period of time sufficient to allow effective binding. Incubation steps can typically be from about 1 minute to twelve hours, at temperatures of about 20° to 30° C., or can be incubated overnight at about 0° C. to about 10° C.

[0099] Following all incubation steps in an ELISA, the contacted surface can be washed so as to remove non-complexed material. A washing procedure can include washing with a solution such as PBS/Tween or borate buffer. Following the formation of specific immunocomplexes between the test sample and the originally bound material, and subsequent washing, the occurrence of even minute amounts of immunocomplexes can be determined.

[0100] To provide a detecting means, the second or third antibody can have an associated label to allow detection, as described above. This can be an enzyme that can generate color development upon incubating with an appropriate chromogenic substrate. Thus, for example, one can contact and incubate the first or second immunocomplex with a labeled antibody for a period of time and under conditions that favor the development of further immunocomplex formation (e.g., incubation for 2 hours at room temperature in a PBS-containing solution such as PBS-Tween).

[0101] After incubation with the labeled antibody, and subsequent to washing to remove unbound material, the amount of label can be quantified, e.g., by incubation with a chromogenic substrate such as urea and bromocresol purple or 2,2'-azido-di-(3-ethyl-benzthiazoline-6-sulfonic acid [ABTS] and H₂O₂, in the case of peroxidase as the enzyme label. Quantitation can then be achieved by measuring the degree of color generation, e.g., using a visible spectra spectrophotometer.

[0102] Protein arrays are solid-phase ligand binding assay systems using immobilized proteins on surfaces which include glass, membranes, microtiter wells, mass spectrometer plates, and beads or other particles. The assays are highly parallel (multiplexed) and often miniaturized (microarrays, protein chips). Their advantages include being rapid and automatable, capable of high sensitivity, economical on reagents, and giving an abundance of data for a single experiment. Bioinformatics support is important; the data

handling demands sophisticated software and data comparison analysis. However, the software can be adapted from that used for DNA arrays, as can much of the hardware and detection systems.

[0103] One of the chief formats is the capture array, in which ligand-binding reagents, which are usually antibodies but can also be alternative protein scaffolds, peptides or nucleic acid aptamers, are used to detect target molecules in mixtures such as plasma or tissue extracts. In diagnostics, capture arrays can be used to carry out multiple immunoassays in parallel, both testing for several analytes in individual sera for example and testing many serum samples simultaneously. In proteomics, capture arrays are used to quantitate and compare the levels of proteins in different samples in health and disease, i.e. protein expression profiling. Proteins other than specific ligand binders are used in the array format for in vitro functional interaction screens such as protein-protein, protein-DNA, protein-drug, receptor-ligand, enzyme-substrate, etc. The capture reagents themselves are selected and screened against many proteins, which can also be done in a multiplex array format against multiple protein targets.

[0104] For construction of arrays, sources of proteins include cell-based expression systems for recombinant proteins, purification from natural sources, production in vitro by cell-free translation systems, and synthetic methods for peptides. Many of these methods can be automated for high throughput production. For capture arrays and protein function analysis, it is important that proteins should be correctly folded and functional; this is not always the case, e.g. where recombinant proteins are extracted from bacteria under denaturing conditions. Nevertheless, arrays of denatured proteins are useful in screening antibodies for cross-reactivity, identifying autoantibodies and selecting ligand binding proteins.

[0105] Protein arrays have been designed as a miniaturization of familiar immunoassay methods such as ELISA and dot blotting, often utilizing fluorescent readout, and facilitated by robotics and high throughput detection systems to enable multiple assays to be carried out in parallel. Commonly used physical supports include glass slides, silicon, microwells, nitrocellulose or PVDF membranes, and magnetic and other microbeads. While microdrops of protein delivered onto planar surfaces are the most familiar format, alternative architectures include CD centrifugation devices based on developments in microfluidics (Gyros, Monmouth Junction, NJ) and specialised chip designs, such as engineered microchannels in a plate (e.g., The Living Chip™, Biotrove, Woburn, MA) and tiny 3D posts on a silicon surface (Zyomyx, Hayward CA). Particles in suspension can also be used as the basis of arrays, providing they are coded for identification; systems include colour coding for microbeads (Luminex, Austin, TX; Bio-Rad Laboratories) and semiconductor nanocrystals (e.g., QDots™, Quantum Dot, Hayward, CA), and barcoding for beads (UltraPlex™, SmartBead Technologies Ltd, Babraham, Cambridge, UK) and multimetal microrods (e.g., Nanobarcodes™ particles, Nanoplex Technologies, Mountain View, CA). Beads can also be assembled into planar arrays on semiconductor chips (LEAPS technology, BioArray Solutions, Warren, NJ).

[0106] Immobilization of proteins involves both the coupling reagent and the nature of the surface being coupled to. A good protein array support surface is chemically stable

before and after the coupling procedures, allows good spot morphology, displays minimal nonspecific binding, does not contribute a background in detection systems, and is compatible with different detection systems. The immobilization method used are reproducible, applicable to proteins of different properties (size, hydrophilic, hydrophobic), amenable to high throughput and automation, and compatible with retention of fully functional protein activity. Orientation of the surface-bound protein is recognized as an important factor in presenting it to ligand or substrate in an active state; for capture arrays the most efficient binding results are obtained with orientated capture reagents, which generally require site-specific labeling of the protein.

[0107] Both covalent and noncovalent methods of protein immobilization are used and have various pros and cons. Passive adsorption to surfaces is methodologically simple, but allows little quantitative or orientational control; it may or may not alter the functional properties of the protein, and reproducibility and efficiency are variable. Covalent coupling methods provide a stable linkage, can be applied to a range of proteins and have good reproducibility; however, orientation may be variable, chemical derivatization may alter the function of the protein and requires a stable interactive surface. Biological capture methods utilizing a tag on the protein provide a stable linkage and bind the protein specifically and in reproducible orientation, but the biological reagent must first be immobilized adequately and the array may require special handling and have variable stability.

[0108] Several immobilization chemistries and tags have been described for fabrication of protein arrays. Substrates for covalent attachment include glass slides coated with amino- or aldehyde-containing silane reagents. In the Versalinx™ system (Prolinx, Bothell, WA) reversible covalent coupling is achieved by interaction between the protein derivatised with phenyldiboronic acid, and salicylhydroxamic acid immobilized on the support surface. This also has low background binding and low intrinsic fluorescence and allows the immobilized proteins to retain function. Noncovalent binding of unmodified protein occurs within porous structures such as HydroGel™ (PerkinElmer, Wellesley, MA), based on a 3-dimensional polyacrylamide gel; this substrate is reported to give a particularly low background on glass microarrays, with a high capacity and retention of protein function. Widely used biological coupling methods are through biotin/streptavidin or hexahistidine/Ni interactions, having modified the protein appropriately. Biotin may be conjugated to a poly-lysine backbone immobilised on a surface such as titanium dioxide (Zyomyx) or tantalum pentoxide (Zeptosens, Witterswil, Switzerland).

[0109] Array fabrication methods include robotic contact printing, ink-jetting, piezoelectric spotting and photolithography. A number of commercial arrayers are available [e.g. Packard Biosciences] as well as manual equipment [V & P Scientific]. Bacterial colonies can be robotically gridded onto PVDF membranes for induction of protein expression in situ.

[0110] At the limit of spot size and density are nanoarrays, with spots on the nanometer spatial scale, enabling thousands of reactions to be performed on a single chip less than 1 mm square. BioForce Laboratories have developed nanoarrays with 1521 protein spots in 85 sq microns, equivalent

to 25 million spots per sq cm, at the limit for optical detection; their readout methods are fluorescence and atomic force microscopy (AFM).

[0111] Fluorescence labeling and detection methods are widely used. The same instrumentation as used for reading DNA microarrays is applicable to protein arrays. For differential display, capture (e.g., antibody) arrays can be probed with fluorescently labeled proteins from two different cell states, in which cell lysates are directly conjugated with different fluorophores (e.g. Cy-3, Cy-5) and mixed, such that the color acts as a readout for changes in target abundance. Fluorescent readout sensitivity can be amplified 10-100 fold by tyramide signal amplification (TSA) (PerkinElmer Lifesciences). Planar waveguide technology (Zeptosens) enables ultrasensitive fluorescence detection, with the additional advantage of no intervening washing procedures. High sensitivity can also be achieved with suspension beads and particles, using phycoerythrin as label (Luminex) or the properties of semiconductor nanocrystals (Quantum Dot). A number of novel alternative readouts have been developed, especially in the commercial biotech arena. These include adaptations of surface plasmon resonance (HTS Biosystems, Intrinsic Bioprobes, Tempe, AZ), rolling circle DNA amplification (Molecular Staging, New Haven CT), mass spectrometry (Intrinsic Bioprobes; Ciphergen, Fremont, CA), resonance light scattering (Genicon Sciences, San Diego, CA) and atomic force microscopy [BioForce Laboratories].

[0112] Capture arrays form the basis of diagnostic chips and arrays for expression profiling. They employ high affinity capture reagents, such as conventional antibodies, single domains, engineered scaffolds, peptides or nucleic acid aptamers, to bind and detect specific target ligands in high throughput manner.

[0113] Antibody arrays have the required properties of specificity and acceptable background, and some are available commercially (BD Biosciences, San Jose, CA; Clontech, Mountain View, CA; BioRad; Sigma, St. Louis, MO). Antibodies for capture arrays are made either by conventional immunization (polyclonal sera and hybridomas), or as recombinant fragments, usually expressed in *E. coli*, after selection from phage or ribosome display libraries (Cambridge Antibody Technology, Cambridge, UK; BioInvent, Lund, Sweden; Affitech, Walnut Creek, CA; Biosite, San Diego, CA). In addition to the conventional antibodies, Fab and scFv fragments, single V-domains from camelids or engineered human equivalents (Domantis, Waltham, MA) may also be useful in arrays.

[0114] The term “scaffold” refers to ligand-binding domains of proteins, which are engineered into multiple variants capable of binding diverse target molecules with antibody-like properties of specificity and affinity. The variants can be produced in a genetic library format and selected against individual targets by phage, bacterial or ribosome display. Such ligand-binding scaffolds or frameworks include ‘Affibodies’ based on *Staph. aureus* protein A (Affibody, Bromma, Sweden), ‘Trinectins’ based on fibronectins (Phylos, Lexington, MA) and ‘Anticalins’ based on the lipocalin structure (Pieris Proteolab, Freising-Weihenstephan, Germany). These can be used on capture arrays in a similar fashion to antibodies and may have advantages of robustness and ease of production.

[0115] Nonprotein capture molecules, notably the single-stranded nucleic acid aptamers which bind protein ligands

with high specificity and affinity, are also used in arrays (SomaLogic, Boulder, CO). Aptamers are selected from libraries of oligonucleotides by the Selex™ procedure and their interaction with protein can be enhanced by covalent attachment, through incorporation of brominated deoxyuridine and UV-activated crosslinking (photoaptamers). Photocrosslinking to ligand reduces the crossreactivity of aptamers due to the specific steric requirements. Aptamers have the advantages of ease of production by automated oligonucleotide synthesis and the stability and robustness of DNA; on photoaptamer arrays, universal fluorescent protein stains can be used to detect binding.

[0116] Protein analytes binding to antibody arrays may be detected directly or via a secondary antibody in a sandwich assay. Direct labelling is used for comparison of different samples with different colours. Where pairs of antibodies directed at the same protein ligand are available, sandwich immunoassays provide high specificity and sensitivity and are therefore the method of choice for low abundance proteins such as cytokines; they also give the possibility of detection of protein modifications. Label-free detection methods, including mass spectrometry, surface plasmon resonance and atomic force microscopy, avoid alteration of ligand. What is required from any method is optimal sensitivity and specificity, with low background to give high signal to noise. Since analyte concentrations cover a wide range, sensitivity has to be tailored appropriately; serial dilution of the sample or use of antibodies of different affinities are solutions to this problem. Proteins of interest are frequently those in low concentration in body fluids and extracts, requiring detection in the pg range or lower, such as cytokines or the low expression products in cells.

[0117] An alternative to an array of capture molecules is one made through ‘molecular imprinting’ technology, in which peptides (e.g., from the C-terminal regions of proteins) are used as templates to generate structurally complementary, sequence-specific cavities in a polymerizable matrix; the cavities can then specifically capture (denatured) proteins that have the appropriate primary amino acid sequence (ProteinPrint™, Aspira Biosystems, Burlingame, CA).

[0118] Another methodology which can be used diagnostically and in expression profiling is the ProteinChip® array (CiphaGen, Fremont, CA), in which solid phase chromatographic surfaces bind proteins with similar characteristics of charge or hydrophobicity from mixtures such as plasma or tumour extracts, and SELDI-TOF mass spectrometry is used to detect the retained proteins.

[0119] Large-scale functional chips have been constructed by immobilizing large numbers of purified proteins and used to assay a wide range of biochemical functions, such as protein interactions with other proteins, drug-target interactions, enzyme-substrates, etc. Generally they require an expression library, cloned into *E. coli*, yeast or similar from which the expressed proteins are then purified, e.g. via a His tag, and immobilized. Cell free protein transcription/translation is a viable alternative for synthesis of proteins which do not express well in bacterial or other in vivo systems.

[0120] For detecting protein-protein interactions, protein arrays can be in vitro alternatives to the cell-based yeast two-hybrid system and may be useful where the latter is deficient, such as interactions involving secreted proteins or proteins with disulphide bridges. High-throughput analysis of biochemical activities on arrays has been described for

yeast protein kinases and for various functions (protein-protein and protein-lipid interactions) of the yeast proteome, where a large proportion of all yeast open-reading frames was expressed and immobilised on a microarray. Large-scale ‘proteome chips’ promise to be very useful in identification of functional interactions, drug screening, etc. (Proteome-trix, Branford, CT).

[0121] As a two-dimensional display of individual elements, a protein array can be used to screen phage or ribosome display libraries, in order to select specific binding partners, including antibodies, synthetic scaffolds, peptides and aptamers. In this way, ‘library against library’ screening can be carried out. Screening of drug candidates in combinatorial chemical libraries against an array of protein targets identified from genome projects is another application of the approach.

[0122] A multiplexed bead assay, such as, for example, the BD™ Cytometric Bead Array, is a series of spectrally discrete particles that can be used to capture and quantitate soluble analytes. The analyte is then measured by detection of a fluorescence-based emission and flow cytometric analysis. Multiplexed bead assay generates data that is comparable to ELISA based assays, but in a “multiplexed” or simultaneous fashion. Concentration of unknowns is calculated for the cytometric bead array as with any sandwich format assay, i.e. through the use of known standards and plotting unknowns against a standard curve. Further, multiplexed bead assay allows quantification of soluble analytes in samples never previously considered due to sample volume limitations. In addition to the quantitative data, powerful visual images can be generated revealing unique profiles or signatures that provide the user with additional information at a glance.

4. Antibodies

(1) Antibodies Generally

[0123] The term “antibodies” is used herein in a broad sense and includes both polyclonal and monoclonal antibodies. In addition to intact immunoglobulin molecules, also included in the term “antibodies” are fragments or polymers of those immunoglobulin molecules, and human or humanized versions of immunoglobulin molecules or fragments thereof, as long as they are chosen for their ability to interact with syndecan binding protein (SDCBP). The antibodies can be tested for their desired activity using the in vitro assays described herein, or by analogous methods, after which their in vivo therapeutic and/or prophylactic activities are tested according to known clinical testing methods. There are five major classes of human immunoglobulins: IgA, IgD, IgE, IgG and IgM, and several of these may be further divided into subclasses (isotypes), e.g., IgG-1, IgG-2, IgG-3, and IgG-4; IgA-1 and IgA-2. One skilled in the art would recognize the comparable classes for mouse. The heavy chain constant domains that correspond to the different classes of immunoglobulins are called alpha, delta, epsilon, gamma, and mu, respectively.

[0124] The term “monoclonal antibody” as used herein refers to an antibody obtained from a substantially homogeneous population of antibodies, i.e., the individual antibodies within the population are identical except for possible naturally occurring mutations that may be present in a small subset of the antibody molecules. The monoclonal antibodies herein specifically include “chimeric” antibodies in

which a portion of the heavy and/or light chain is identical with or homologous to corresponding sequences in antibodies derived from a particular species or belonging to a particular antibody class or subclass, while the remainder of the chain(s) is identical with or homologous to corresponding sequences in antibodies derived from another species or belonging to another antibody class or subclass, as well as fragments of such antibodies, as long as they exhibit the desired antagonistic activity.

[0125] The disclosed monoclonal antibodies can be made using any procedure which produces monoclonal antibodies. For example, disclosed monoclonal antibodies can be prepared using hybridoma methods, such as those described by Kohler and Milstein, *Nature*, 256:495 (1975). In a hybridoma method, a mouse or other appropriate host animal is typically immunized with an immunizing agent to elicit lymphocytes that produce or are capable of producing antibodies that will specifically bind to the immunizing agent. Alternatively, the lymphocytes may be immunized in vitro.

[0126] The monoclonal antibodies may also be made by recombinant DNA methods. DNA encoding the disclosed monoclonal antibodies can be readily isolated and sequenced using conventional procedures (e.g., by using oligonucleotide probes that are capable of binding specifically to genes encoding the heavy and light chains of murine antibodies). Libraries of antibodies or active antibody fragments can also be generated and screened using phage display techniques, e.g., as described in U.S. Pat. No. 5,804,440 to Burton et al. and U.S. Pat. No. 6,096,441 to Barbas et al.

[0127] In vitro methods are also suitable for preparing monovalent antibodies. Digestion of antibodies to produce fragments thereof, particularly, Fab fragments, can be accomplished using routine techniques known in the art. For instance, digestion can be performed using papain. Examples of papain digestion are described in WO 94/29348 published Dec. 22, 1994 and U.S. Pat. No. 4,342,566. Papain digestion of antibodies typically produces two identical antigen binding fragments, called Fab fragments, each with a single antigen binding site, and a residual Fc fragment. Pepsin treatment yields a fragment that has two antigen combining sites and is still capable of cross-linking antigen.

[0128] As used herein, the term “antibody or fragments thereof” encompasses chimeric antibodies and hybrid antibodies, with dual or multiple antigen or epitope specificities, and fragments, such as F(ab')₂, Fab', Fab, Fv, sFv, scFv, and the like, including hybrid fragments. Thus, fragments of the antibodies that retain the ability to bind their specific antigens are provided. For example, fragments of antibodies which maintain syndecan binding protein (SDCBP) binding activity are included within the meaning of the term “antibody or fragment thereof.” Such antibodies and fragments can be made by techniques known in the art and can be screened for specificity and activity according to the methods set forth in the Examples and in general methods for producing antibodies and screening antibodies for specificity and activity (See Harlow and Lane, *Antibodies, A Laboratory Manual*. Cold Spring Harbor Publications, New York, (1988)).

[0129] Also included within the meaning of “antibody or fragments thereof” are conjugates of antibody fragments and antigen binding proteins (single chain antibodies).

[0130] The fragments, whether attached to other sequences or not, can also include insertions, deletions, substitutions, or other selected modifications of particular regions or specific amino acids residues, provided the activity of the antibody or antibody fragment is not significantly altered or impaired compared to the non-modified antibody or antibody fragment. These modifications can provide for some additional property, such as to remove/add amino acids capable of disulfide bonding, to increase its bio-longevity, to alter its secretory characteristics, etc. In any case, the antibody or antibody fragment must possess a bioactive property, such as specific binding to its cognate antigen. Functional or active regions of the antibody or antibody fragment may be identified by mutagenesis of a specific region of the protein, followed by expression and testing of the expressed polypeptide. Such methods are readily apparent to a skilled practitioner in the art and can include site-specific mutagenesis of the nucleic acid encoding the antibody or antibody fragment. (Zoller, M. J. *Curr. Opin. Biotechnol.* 3:348-354, 1992).

[0131] As used herein, the term “antibody” or “antibodies” can also refer to a human antibody and/or a humanized antibody. Many non-human antibodies (e.g., those derived from mice, rats, or rabbits) are naturally antigenic in humans, and thus can give rise to undesirable immune responses when administered to humans. Therefore, the use of human or humanized antibodies in the methods serves to lessen the chance that an antibody administered to a human will evoke an undesirable immune response.

(2) Human Antibodies

[0132] The disclosed human antibodies can be prepared using any technique. The disclosed human antibodies can also be obtained from transgenic animals. For example, transgenic, mutant mice that are capable of producing a full repertoire of human antibodies, in response to immunization, have been described (see, e.g., Jakobovits et al., *Proc. Natl. Acad. Sci. USA*, 90:2551-255 (1993); Jakobovits et al., *Nature*, 362:255-258 (1993); Bruggermann et al., *Year in Immunol.*, 7:33 (1993)). Specifically, the homozygous deletion of the antibody heavy chain joining region (J(H)) gene in these chimeric and germ-line mutant mice results in complete inhibition of endogenous antibody production, and the successful transfer of the human germ-line antibody gene array into such germ-line mutant mice results in the production of human antibodies upon antigen challenge. Antibodies having the desired activity are selected using Env-CD4-co-receptor complexes as described herein.

(3) Humanized Antibodies

[0133] Antibody humanization techniques generally involve the use of recombinant DNA technology to manipulate the DNA sequence encoding one or more polypeptide chains of an antibody molecule. Accordingly, a humanized form of a non-human antibody (or a fragment thereof) is a chimeric antibody or antibody chain (or a fragment thereof, such as an sFv, Fv, Fab, Fab', F(ab')₂, or other antigen-binding portion of an antibody) which contains a portion of an antigen binding site from a non-human (donor) antibody integrated into the framework of a human (recipient) antibody.

[0134] To generate a humanized antibody, residues from one or more complementarity determining regions (CDRs)

of a recipient (human) antibody molecule are replaced by residues from one or more CDRs of a donor (non-human) antibody molecule that is known to have desired antigen binding characteristics (e.g., a certain level of specificity and affinity for the target antigen). In some instances, Fv framework (FR) residues of the human antibody are replaced by corresponding non-human residues. Humanized antibodies may also contain residues which are found neither in the recipient antibody nor in the imported CDR or framework sequences. Generally, a humanized antibody has one or more amino acid residues introduced into it from a source which is non-human. In practice, humanized antibodies are typically human antibodies in which some CDR residues and possibly some FR residues are substituted by residues from analogous sites in rodent antibodies. Humanized antibodies generally contain at least a portion of an antibody constant region (Fc), typically that of a human antibody (Jones et al., *Nature*, 321:522-525 (1986), Reichmann et al., *Nature*, 332:323-327 (1988), and Presta, *Curr. Opin. Struct. Biol.*, 2:593-596 (1992)).

[0135] Methods for humanizing non-human antibodies are well known in the art. For example, humanized antibodies can be generated according to the methods of Winter and co-workers (Jones et al., *Nature*, 321:522-525 (1986), Riechmann et al., *Nature*, 332:323-327 (1988), Verhoeyen et al., *Science*, 239:1534-1536 (1988)), by substituting rodent CDRs or CDR sequences for the corresponding sequences of a human antibody. Methods that can be used to produce humanized antibodies are also described in U.S. Pat. No. 4,816,567 (Cabilly et al.), U.S. Pat. No. 5,565,332 (Hoo-genboom et al.), U.S. Pat. No. 5,721,367 (Kay et al.), U.S. Pat. No. 5,837,243 (Deo et al.), U.S. Pat. No. 5,939,598 (Kucherlapati et al.), U.S. Pat. No. 6,130,364 (Jakobovits et al.), and U.S. Pat. No. 6,180,377 (Morgan et al.).

(4) Administration of Antibodies

[0136] Administration of the antibodies can be done as disclosed herein. Nucleic acid approaches for antibody delivery also exist. The broadly neutralizing anti-SDCBP antibodies and antibody fragments can also be administered to patients or subjects as a nucleic acid preparation (e.g., DNA or RNA) that encodes the antibody or antibody fragment, such that the patient's or subject's own cells take up the nucleic acid and produce and secrete the encoded antibody or antibody fragment. The delivery of the nucleic acid can be by any means, as disclosed herein, for example.

5. Pharmaceutical Carriers/Delivery of Pharmaceutical Products

[0137] As described above, the compositions can also be administered in vivo in a pharmaceutically acceptable carrier. By "pharmaceutically acceptable" is meant a material that is not biologically or otherwise undesirable, i.e., the material may be administered to a subject, along with the nucleic acid or vector, without causing any undesirable biological effects or interacting in a deleterious manner with any of the other components of the pharmaceutical composition in which it is contained. The carrier would naturally be selected to minimize any degradation of the active ingredient and to minimize any adverse side effects in the subject, as would be well known to one of skill in the art.

[0138] The compositions may be administered orally, parenterally (e.g., intravenously), by intramuscular injection, by

intraperitoneal injection, transdermally, extracorporeally, topically or the like, including topical intranasal administration or administration by inhalant. As used herein, "topical intranasal administration" means delivery of the compositions into the nose and nasal passages through one or both of the nares and can comprise delivery by a spraying mechanism or droplet mechanism, or through aerosolization of the nucleic acid or vector. Administration of the compositions by inhalant can be through the nose or mouth via delivery by a spraying or droplet mechanism. Delivery can also be directly to any area of the respiratory system (e.g., lungs) via intubation. The exact amount of the compositions required will vary from subject to subject, depending on the species, age, weight and general condition of the subject, the severity of the allergic disorder being treated, the particular nucleic acid or vector used, its mode of administration and the like. Thus, it is not possible to specify an exact amount for every composition. However, an appropriate amount can be determined by one of ordinary skill in the art using only routine experimentation given the teachings herein.

[0139] Parenteral administration of the composition, if used, is generally characterized by injection. Injectables can be prepared in conventional forms, either as liquid solutions or suspensions, solid forms suitable for solution of suspension in liquid prior to injection, or as emulsions. A more recently revised approach for parenteral administration involves use of a slow release or sustained release system such that a constant dosage is maintained. See, e.g., U.S. Pat. No. 3,610,795, which is incorporated by reference herein.

[0140] The materials may be in solution, suspension (for example, incorporated into microparticles, liposomes, or cells). These may be targeted to a particular cell type via antibodies, receptors, or receptor ligands. The following references are examples of the use of this technology to target specific proteins to tumor tissue (Senter, et al., *Bioconjugate Chem.*, 2:447-451, (1991); Bagshawe, K. D., *Br. J. Cancer*, 60:275-281, (1989); Bagshawe, et al., *Br. J. Cancer*, 58:700-703, (1988); Senter, et al., *Bioconjugate Chem.*, 4:3-9, (1993); Battelli, et al., *Cancer Immunol. Immunother.*, 35:421-425, (1992); Pietersz and McKenzie, *Immunolog. Reviews*, 129:57-80, (1992); and Roffler, et al., *Biochem. Pharmacol.*, 42:2062-2065, (1991)). Vehicles such as "stealth" and other antibody conjugated liposomes (including lipid mediated drug targeting to colonic carcinoma), receptor mediated targeting of DNA through cell specific ligands, lymphocyte directed tumor targeting, and highly specific therapeutic retroviral targeting of murine glioma cells in vivo. The following references are examples of the use of this technology to target specific proteins to tumor tissue (Hughes et al., *Cancer Research*, 49:6214-6220, (1989); and Litzinger and Huang, *Biochimica et Biophysica Acta*, 1104:179-187, (1992)). In general, receptors are involved in pathways of endocytosis, either constitutive or ligand induced. These receptors cluster in clathrin-coated pits, enter the cell via clathrin-coated vesicles, pass through an acidified endosome in which the receptors are sorted, and then either recycle to the cell surface, become stored intracellularly, or are degraded in lysosomes. The internalization pathways serve a variety of functions, such as nutrient uptake, removal of activated proteins, clearance of macromolecules, opportunistic entry of viruses and toxins, dissociation and degradation of ligand, and receptor-level regulation. Many receptors follow more than one intracellular pathway, depending on the cell type, receptor concentration,

type of ligand, ligand valency, and ligand concentration. Molecular and cellular mechanisms of receptor-mediated endocytosis has been reviewed (Brown and Greene, DNA and Cell Biology 10:6, 399-409 (1991)).

a) Pharmaceutically Acceptable Carriers

[0141] The compositions, including antibodies, can be used therapeutically in combination with a pharmaceutically acceptable carrier.

[0142] Suitable carriers and their formulations are described in *Remington: The Science and Practice of Pharmacy* (19th ed.) ed. A. R. Gennaro, Mack Publishing Company, Easton, PA 1995. Typically, an appropriate amount of a pharmaceutically-acceptable salt is used in the formulation to render the formulation isotonic. Examples of the pharmaceutically-acceptable carrier include, but are not limited to, saline, Ringer's solution and dextrose solution. The pH of the solution is preferably from about 5 to about 8, and more preferably from about 7 to about 7.5. Further carriers include sustained release preparations such as semipermeable matrices of solid hydrophobic polymers containing the antibody, which matrices are in the form of shaped articles, e.g., films, liposomes or microparticles. It will be apparent to those persons skilled in the art that certain carriers may be more preferable depending upon, for instance, the route of administration and concentration of composition being administered.

[0143] Pharmaceutical carriers are known to those skilled in the art. These most typically would be standard carriers for administration of drugs to humans, including solutions such as sterile water, saline, and buffered solutions at physiological pH. The compositions can be administered intramuscularly or subcutaneously. Other compounds will be administered according to standard procedures used by those skilled in the art.

[0144] Pharmaceutical compositions may include carriers, thickeners, diluents, buffers, preservatives, surface active agents and the like in addition to the molecule of choice. Pharmaceutical compositions may also include one or more active ingredients such as antimicrobial agents, antiinflammatory agents, anesthetics, and the like.

[0145] The pharmaceutical composition may be administered in a number of ways depending on whether local or systemic treatment is desired, and on the area to be treated. Administration may be topically (including ophthalmically, vaginally, rectally, intranasally), orally, by inhalation, or parenterally, for example by intravenous drip, subcutaneous, intraperitoneal or intramuscular injection. The disclosed antibodies can be administered intravenously, intraperitoneally, intramuscularly, subcutaneously, intracavity, or transdermally.

[0146] Preparations for parenteral administration include sterile aqueous or non-aqueous solutions, suspensions, and emulsions. Examples of non-aqueous solvents are propylene glycol, polyethylene glycol, vegetable oils such as olive oil, and injectable organic esters such as ethyl oleate. Aqueous carriers include water, alcoholic/aqueous solutions, emulsions or suspensions, including saline and buffered media. Parenteral vehicles include sodium chloride solution, Ringer's dextrose, dextrose and sodium chloride, lactated Ringer's, or fixed oils. Intravenous vehicles include fluid and nutrient replenishers, electrolyte replenishers (such as those based on Ringer's dextrose), and the like. Preservatives and

other additives may also be present such as, for example, antimicrobials, anti-oxidants, chelating agents, and inert gases and the like.

[0147] Formulations for topical administration may include ointments, lotions, creams, gels, drops, suppositories, sprays, liquids and powders. Conventional pharmaceutical carriers, aqueous, powder or oily bases, thickeners and the like may be necessary or desirable.

[0148] Compositions for oral administration include powders or granules, suspensions or solutions in water or non-aqueous media, capsules, sachets, or tablets. Thickeners, flavorings, diluents, emulsifiers, dispersing aids or binders may be desirable.

[0149] Some of the compositions may potentially be administered as a pharmaceutically acceptable acid- or base-addition salt, formed by reaction with inorganic acids such as hydrochloric acid, hydrobromic acid, perchloric acid, nitric acid, thiocyanic acid, sulfuric acid, and phosphoric acid, and organic acids such as formic acid, acetic acid, propionic acid, glycolic acid, lactic acid, pyruvic acid, oxalic acid, malonic acid, succinic acid, maleic acid, and fumaric acid, or by reaction with an inorganic base such as sodium hydroxide, ammonium hydroxide, potassium hydroxide, and organic bases such as mono-, di-, trialkyl and aryl amines and substituted ethanolamines.

b) Therapeutic Uses

[0150] Effective dosages and schedules for administering the compositions may be determined empirically, and making such determinations is within the skill in the art. The dosage ranges for the administration of the compositions are those large enough to produce the desired effect in which the symptoms of the disorder are effected. The dosage should not be so large as to cause adverse side effects, such as unwanted cross-reactions, anaphylactic reactions, and the like. Generally, the dosage will vary with the age, condition, sex and extent of the disease in the patient, route of administration, or whether other drugs are included in the regimen, and can be determined by one of skill in the art. The dosage can be adjusted by the individual physician in the event of any counterindications. Dosage can vary, and can be administered in one or more dose administrations daily, for one or several days. Guidance can be found in the literature for appropriate dosages for given classes of pharmaceutical products. For example, guidance in selecting appropriate doses for antibodies can be found in the literature on therapeutic uses of antibodies, e.g., *Handbook of Monoclonal Antibodies*, Ferrone et al., eds., Noyes Publications, Park Ridge, N.J., (1985) ch. 22 and pp. 303-357; Smith et al., *Antibodies in Human Diagnosis and Therapy*, Haber et al., eds., Raven Press, New York (1977) pp. 365-389. A typical daily dosage of the antibody used alone might range from about 1 µg/kg to up to 100 mg/kg of body weight or more per day, depending on the factors mentioned above.

C. METHODS OF USING THE COMPOSITIONS

1. Method of Treating Cancer

[0151] The disclosed compositions can be used to treat any disease where uncontrolled cellular proliferation occurs such as cancers. A representative but non-limiting list of cancers that the disclosed compositions can be used to treat

is the following: lymphomas such as B cell lymphoma and T cell lymphoma; mycosis fungoides; Hodgkin's Disease; myeloid leukemia (including, but not limited to acute myeloid leukemia (AML) and/or chronic myeloid leukemia (CML)); bladder cancer; brain cancer; nervous system cancer; head and neck cancer; squamous cell carcinoma of head and neck; renal cancer; lung cancers such as small cell lung cancer, non-small cell lung carcinoma (NSCLC), lung squamous cell carcinoma (LUSC), and Lung Adenocarcinomas (LUAD); neuroblastoma/glioblastoma; ovarian cancer (including, but not limited to clear cell carcinoma (CCC), endometrioid carcinoma (EC), high-grade serous ovarian carcinoma (HGSOC), and endometriosis-associated ovarian cancer (EOAC)); pancreatic cancer; prostate cancer; skin cancer; hepatic cancer; melanoma; squamous cell carcinomas of the mouth, throat, larynx, and lung; cervical cancer; cervical carcinoma; breast cancer including, but not limited to triple negative breast cancer; genitourinary cancer; pulmonary cancer; esophageal carcinoma; head and neck carcinoma; large bowel cancer; hematopoietic cancers; testicular cancer; and colon and rectal cancers.

[0152] In one aspect, the treatment of the cancer can include administration of an anti-SDCBP binding molecule. For example, disclosed herein are methods of treating, inhibiting, reducing, decreasing, ameliorating, and/or preventing a cancer and/or metastasis (such as, for example, ovarian cancer, including, but not limited to clear cell carcinoma (CCC), endometrioid carcinoma (EC), high-grade serous ovarian carcinoma (HGSOC), and endometriosis-associated ovarian cancer (EOAC)) or endometriosis in a subject comprising administering to the subject an anti-SDCBP binding molecule as set forth herein. Accordingly, in one aspect, disclosed herein are methods of treating, inhibiting, reducing, decreasing, ameliorating, and/or preventing a cancer and/or metastasis (such as, for example, ovarian cancer including, but not limited to clear cell carcinoma (CCC), endometrioid carcinoma (EC), high-grade serous ovarian carcinoma (HGSOC), or endometriosis-associated ovarian cancer (EOAC)) in a subject comprising administering to the subject any of the anti-SDCBP binding molecules disclosed herein. For example, disclosed herein are methods treating, inhibiting, reducing, decreasing, ameliorating, and/or preventing a cancer and/or metastasis (such as, for example, ovarian cancer including, but not limited to clear cell carcinoma (CCC), endometrioid carcinoma (EC), high-grade serous ovarian carcinoma (HGSOC), or endometriosis-associated ovarian cancer (EOAC)) or endometriosis in a subject comprising administering to the subject an anti-syndecan binding protein (SDCBP) binding molecule comprising a heavy chain variable domain complementary determining region (CDR) 1 (CDR1), CDR2, and CDR3 as set forth in SEQ ID NO: 7, SEQ ID NO: 8, and SEQ ID NO: 9, respectively (such as, for example an anti-SDCBP binding molecule comprising a heavy chain variable domain comprising the amino acid sequence as set forth in SEQ ID NO: 6); SEQ ID NO: 7, SEQ ID NO: 8, and SEQ ID NO: 15, respectively; SEQ ID NO: 7, SEQ ID NO: 8, and SEQ ID NO: 16, respectively; or SEQ ID NO: 7, SEQ ID NO: 8, and SEQ ID NO: 17, respectively. In one aspect the heavy chain variable domain comprises the amino acid sequence as set forth in SEQ ID NO: 18. In one aspect, the anti-SDCBP binding molecule further comprises a light chain variable domain complementary determining region (CDR) 1 (CDR1), CDR2, and

CDR3 as set forth in SEQ ID NO: 11, SEQ ID NO: 12, and SEQ ID NO: 13, respectively (such as, for example an anti-SDCBP binding molecule comprising a light chain variable domain comprising the amino acid sequence as set forth in SEQ ID NO: 10) or SEQ ID NO: 11, SEQ ID NO: 12, and SEQ ID NO: 14, respectively. In one aspect the light chain variable domain comprises the amino acid sequence as set forth in SEQ ID NO: 198.

[0153] It is understood and herein contemplated that the disclosed treatment regimens can be used alone or in combination with any anti-cancer therapy known in the art including, but not limited to Abemaciclib, Abiraterone Acetate, Abitrexate (Methotrexate), Abraxane (Paclitaxel Albumin-stabilized Nanoparticle Formulation), ABVD, ABVE, ABVE-PC, AC, AC-T, Adcetris (Brentuximab Vedotin), ADE, Ado-Trastuzumab Emtansine, Adriamycin (Doxorubicin Hydrochloride), Afatinib Dimaleate, Afinitor (Everolimus), Akynzeo (Netupitant and Palonosetron Hydrochloride), Aldara (Imiquimod), Aldesleukin, Alecensa (Alectinib), Alectinib, Alemtuzumab, Alimta (Pemetrexed Disodium), Aliqopa (Copanlisib Hydrochloride), Alkeran for Injection (Melphalan Hydrochloride), Alkeran Tablets (Melphalan), Aloxi (Palonosetron Hydrochloride), Alunbrig (Brigatinib), Ambochlorin (Chlorambucil), Amboclorin Chlorambucil, Amifostine, Aminolevulinic Acid, Anastrozole, Aprepitant, Aredia (Pamidronate Disodium), Arimidex (Anastrozole), Aromasin (Exemestane), Arranon (Nelarabine), Arsenic Trioxide, Arzerra (Ofatumumab), Asparaginase *Erwinia chrysanthemi*, Atezolizumab, Avastin (Bevacizumab), Avelumab, Axitinib, Azacitidine, Bavencio (Avelumab), BEACOPP, Becenun (Carmustine), Beleodaq (Belinostat), Belinostat, Bendamustine Hydrochloride, BEP, Besponsa (Inotuzumab Ozogamicin), Bevacizumab, Bexarotene, Bexxar (Tositumomab and Iodine I 131 Tositumomab), Bicalutamide, BiCNU (Carmustine), Bleomycin, Blinatumomab, Blincyto (Blinatumomab), Bortezomib, Bosulif (Bosutinib), Bosutinib, Brentuximab Vedotin, Brigatinib, BuMel, Busulfan, Busulfex (Busulfan), Cabazitaxel, Cabometyx (Cabozantinib-S-Malate), Cabozantinib-S-Malate, CAF, Campath (Alemtuzumab), Camptosar (Irinotecan Hydrochloride), Capecitabine, CAPOX, Carac (Fluorouracil-Topical), Carboplatin, CARBOPLATIN-TAXOL, Carfilzomib, Carmubris (Carmustine), Carmustine, Carmustine Implant, Casodex (Bicalutamide), CEM, Ceritinib, Cerubidine (Daunorubicin Hydrochloride), Cervarix (Recombinant HPV Bivalent Vaccine), Cetuximab, CEV, Chlorambucil, CHLORAMBUCIL-PREDNISONE, CHOP, Cisplatin, Cladribine, Clafen (Cyclophosphamide), Clofarabine, Clofarex (Clofarabine), Clolar (Clofarabine), CMF, Cobimetinib, Cometriq (Cabozantinib-S-Malate), Copanlisib Hydrochloride, COPDAC, COPP, COPP-ABV, Cosmegen (Dactinomycin), Cotellic (Cobimetinib), Crizotinib, CVP, Cyclophosphamide, Cyfos (Ifosfamide), Cyramza (Ramucirumab), Cytarabine, Cytarabine Liposome, Cytosar-U (Cytarabine), Cytoxan (Cyclophosphamide), Dabrafenib, Dacarbazine, Dacogen (Decitabine), Dactinomycin, Daratumumab, Darzalex (Daratumumab), Dasatinib, Daunorubicin Hydrochloride, Daunorubicin Hydrochloride and Cytarabine Liposome, Decitabine, Defibrotide Sodium, Defitelio (Defibrotide Sodium), Degarelix, Denileukin Diftitox, Denosumab, DepoCyt (Cytarabine Liposome), Dexamethasone, Dexrazoxane Hydrochloride, Dinutuximab, Docetaxel, Doxil (Doxorubicin Hydrochloride Liposome), Doxorubicin Hydrochloride, Doxorubicin Hydrochloride

Liposome, Dox-SL (Doxorubicin Hydrochloride Liposome), DTIC-Dome (Dacarbazine), Durvalumab, Efudex (Fluorouracil-Topical), Elitek (Rasburicase), Ellence (Epirubicin Hydrochloride), Elotuzumab, Eloxatin (Oxaliplatin), Eltrombopag Olamine, Emend (Aprepitant), Empliciti (Elotuzumab), Enasidenib Mesylate, Enzalutamide, Epirubicin Hydrochloride, EPOCH, Erbitux (Cetuximab), Eribulin Mesylate, Erivedge (Vismodegib), Erlotinib Hydrochloride, Erwinaze (Asparaginase *Erwinia chrysanthemi*), Ethyol (Amifostine), Etopophos (Etoposide Phosphate), Etoposide, Etoposide Phosphate, Evacet (Doxorubicin Hydrochloride Liposome), Everolimus, Evista, (Raloxifene Hydrochloride), Evomela (Melphalan Hydrochloride), Exemestane, 5-FU (Fluorouracil Injection), 5-FU (Fluorouracil-Topical), Fareston (Toremifene), Farydak (Panobinostat), Faslodex (Fulvestrant), FEC, Femara (Letrozole), Filgrastim, Fludara (Fludarabine Phosphate), Fludarabine Phosphate, Fluoroplex (Fluorouracil-Topical), Fluorouracil Injection, Fluorouracil-Topical, Flutamide, Folex (Methotrexate), Folex PFS (Methotrexate), FOLFIRI, FOLFIRI-BEVACIZUMAB, FOLFIRI-CETUXIMAB, FOLFIRINOX, FOLFOX, Folutyn (Pralatrexate), FU-LV, Fulvestrant, Gardasil (Recombinant HPV Quadrivalent Vaccine), Gardasil 9 (Recombinant HPV Nonavalent Vaccine), Gazyva (Obinutuzumab), Gefitinib, Gemcitabine Hydrochloride, GEMCITABINE-CISPLATIN, GEMCITABINE-OXALIPLATIN, Gemtuzumab Ozogamicin, Gemzar (Gemcitabine Hydrochloride), Giletrif (Afatinib Dimaleate), Gleevec (Imatinib Mesylate), Gliadel (Carmustine Implant), Gliadel wafer (Carmustine Implant), Glucarpidase, Goserelin Acetate, Halaven (Eribulin Mesylate), Hemangeol (Propranolol Hydrochloride), Herceptin (Trastuzumab), HPV Bivalent Vaccine, Recombinant, HPV Nonavalent Vaccine, Recombinant, HPV Quadrivalent Vaccine, Recombinant, Hycamtin (Topotecan Hydrochloride), Hydrea (Hydroxyurea), Hydroxyurea, Hyper-CVAD, Ibrance (Palbociclib), Ibritumomab Tiuxetan, Ibrutinib, ICE, Iclusig (Ponatinib Hydrochloride), Idamycin (Idarubicin Hydrochloride), Idarubicin Hydrochloride, Idelalisib, Idhifa (Enasidenib Mesylate), Ifex (Ifosfamide), Ifosfamide, Ifosfamidum (Ifosfamide), IL-2 (Aldesleukin), Imatinib Mesylate, Imbruvica (Ibrutinib), Imfinzi (Durvalumab), Imiquimod, Imlygic (Talinogene Laherparepvec), Inlyta (Axitinib), Inotuzumab Ozogamicin, Interferon Alfa-2b, Recombinant, Interleukin-2 (Aldesleukin), Intron A (Recombinant Interferon Alfa-2b), Iodine I 131 Tositumomab and Tositumomab, Ipilimumab, Iressa (Gefitinib), Irinotecan Hydrochloride, Irinotecan Hydrochloride Liposome, Istodax (Romidepsin), Ixabepilone, Ixazomib Citrate, Ixempra (Ixabepilone), Jakafi (Ruxolitinib Phosphate), JEB, Jevtana (Cabazitaxel), Kadcyla (Ado-Trastuzumab Emtansine), Keoxifene (Raloxifene Hydrochloride), Kepivance (Palifermin), Keytruda (Pembrolizumab), Kisqali (Ribociclib), Kymriah (Tisagenlecleucel), Kyprolis (Carfilzomib), Lanreotide Acetate, Lapatinib Ditosylate, Lartruvo (Olaratumab), Lenalidomide, Lenvatinib Mesylate, Lenvima (Lenvatinib Mesylate), Letrozole, Leucovorin Calcium, Leukeran (Chlorambucil), Leuprolide Acetate, Leustatin (Cladribine), Levulan (Aminolevulinic Acid), Linfoliniz (Chlorambucil), LipoDox (Doxorubicin Hydrochloride Liposome), Lomustine, Lonsurf (Trifluridine and Tipiracil Hydrochloride), Lupron (Leuprolide Acetate), Lupron Depot (Leuprolide Acetate), Lupron Depot-Ped (Leuprolide Acetate), Lynparza (Olaparib), Marqibo (Vincristine Sulfate

Liposome), Matulane (Procarbazine Hydrochloride), Mechlorethamine Hydrochloride, Megestrol Acetate, Mekinist (Trametinib), Melphalan, Melphalan Hydrochloride, Mercaptopurine, Mesna, Mesnex (Mesna), Methazostone (Temozolomide), Methotrexate, Methotrexate LPF (Methotrexate), Methylnaltrexone Bromide, Mexate (Methotrexate), Mexate-AQ (Methotrexate), Midostaurin, Mitomycin C, Mitoxantrone Hydrochloride, Mitozytrex (Mitomycin C), MOPP, Mozobil (Plerixafor), Mustargen (Mechlorethamine Hydrochloride), Mutamycin (Mitomycin C), Myleran (Busulfan), Mylosar (Azacitidine), Mylotarg (Gemtuzumab Ozogamicin), Nanoparticle Paclitaxel (Paclitaxel Albumin-stabilized Nanoparticle Formulation), Navelbine (Vinorelbine Tartrate), Necitumumab, Nelarabine, Neosar (Cyclophosphamide), Neratinib Maleate, Nerlynx (Neratinib Maleate), Netupitant and Palonosetron Hydrochloride, Neulasta (Pegfilgrastim), Neupogen (Filgrastim), Nexavar (Sorafenib Tosylate), Nilandron (Nilutamide), Nilotinib, Nilutamide, Ninlaro (Ixazomib Citrate), Niraparib Tosylate Monohydrate, Nivolumab, Nolvadex (Tamoxifen Citrate), Nplate (Romiplostim), Obinutuzumab, Odomzo (Sonidegib), OEPA, Ofatumumab, OFF, Olaparib, Olaratumab, Omacetaxine Mepesuccinate, Oncaspar (Pegaspargase), Ondansetron Hydrochloride, Onivyde (Irinotecan Hydrochloride Liposome), Ontak (Denileukin Difitox), Opdivo (Nivolumab), OPPA, Osimertinib, Oxaliplatin, Paclitaxel, Paclitaxel Albumin-stabilized Nanoparticle Formulation, PAD, Palbociclib, Palifermin, Palonosetron Hydrochloride, Palonosetron Hydrochloride and Netupitant, Pamidronate Disodium, Panitumumab, Panobinostat, Paraplat (Carboplatin), Paraplatin (Carboplatin), Pazopanib Hydrochloride, PCV, PEB, Pegaspargase, Pegfilgrastim, Peginterferon Alfa-2b, PEG-Intron (Peginterferon Alfa-2b), Pembrolizumab, Pemetrexed Disodium, Perjeta (Pertuzumab), Pertuzumab, Platinol (Cisplatin), Platinol-AQ (Cisplatin), Plerixafor, Pomalidomide, Pomalyst (Pomalidomide), Ponatinib Hydrochloride, Portrazza (Necitumumab), Pralatrexate, Prednisone, Procarbazine Hydrochloride, Proleukin (Aldesleukin), Prolia (Denosumab), Promacta (Eltrombopag Olamine), Propranolol Hydrochloride, Provenge (Sipuleucel-T), Purinethol (Mercaptopurine), Purixan (Mercaptopurine), Radium 223 Dichloride, Raloxifene Hydrochloride, Ramucirumab, Rasburicase, R-CHOP, R-CVP, Recombinant Human Papillomavirus (HPV) Bivalent Vaccine, Recombinant Human Papillomavirus (HPV) Nonavalent Vaccine, Recombinant Human Papillomavirus (HPV) Quadrivalent Vaccine, Recombinant Interferon Alfa-2b, Regorafenib, Relistor (Methylnaltrexone Bromide), R-EP-OCH, Revlimid (Lenalidomide), Rheumatrex (Methotrexate), Ribociclib, R-ICE, Rituxan (Rituximab), Rituxan Hycela (Rituximab and Hyaluronidase Human), Rituximab, Rituximab and, Hyaluronidase Human, Rolapitant Hydrochloride, Romidepsin, Romiplostim, Rubidomycin (Daunorubicin Hydrochloride), Rubraca (Rucaparib Camsylate), Rucaparib Camsylate, Ruxolitinib Phosphate, Rydapt (Midostaurin), Sclerosol Intrapleural Aerosol (Tale), Siltuximab, Sipuleucel-T, Somatuline Depot (Lanreotide Acetate), Sonidegib, Sorafenib Tosylate, Sprycel (Dasatinib), STANFORD V, Sterile Talc Powder (Talc), Sterilate (Talc), Stivarga (Regorafenib), Sunitinib Malate, Sutent (Sunitinib Malate), Sylatron (Peginterferon Alfa-2b), Sylvant (Siltuximab), Synribo (Omacetaxine Mepesuccinate), Tabloid (Thioguanine), TAC, Tafinlar (Dabrafenib), Tagrisso (Osimertinib), Tale, Talimogene Laherparepvec, Tamoxifen Cit

rate, Tarabine PFS (Cytarabine), Tarceva (Erlotinib Hydrochloride), Targretin (Bexarotene), Tasigna (Nilotinib), Taxol (Paclitaxel), Taxotere (Docetaxel), Tecentriq, (Atezolizumab), Temodar (Temozolomide), Temozolomide, Temsirolimus, Thalidomide, Thalomid (Thalidomide), Thioguanine, Thiotepa, Tisagenlecleucel, Tolak (Fluorouracil-Topical), Topotecan Hydrochloride, Toremifene, Torisel (Temsilolimus), Tositumomab and Iodine I 131 Tositumomab, Totect (Dexrazoxane Hydrochloride), TPF, Trabectedin, Trametinib, Trastuzumab, Treanda (Bendamustine Hydrochloride), Trifluridine and Tipiracil Hydrochloride, Trisenox (Arsenic Trioxide), Tykerb (Lapatinib Ditosylate), Unituxin (Dinutuximab), Uridine Triacetate, VAC, Vandetanib, VAMP, Varubi (Rolapitant Hydrochloride), Vectibix (Panitumumab), Velp, Velban (Vinblastine Sulfate), Velcade (Bortezomib), Velsar (Vinblastine Sulfate), Vemurafenib, Venclexta (Venetoclax), Venetoclax, Verzenio (Abemaciclib), Viadur (Leuprolide Acetate), Vidaza (Azacitidine), Vinblastine Sulfate, Vincasar PFS (Vincristine Sulfate), Vincristine Sulfate, Vincristine Sulfate Liposome, Vinorelbine Tartrate, VIP, Vismodegib, Vistogard (Uridine Triacetate), Voraxaze (Glucarpidase), Vorinostat, Votrient (Pazopanib Hydrochloride), Vyxeos (Daunorubicin Hydrochloride and Cytarabine Liposome), Wellcovorin (Leucovorin Calcium), Xalkori (Crizotinib), Xeloda (Capecitabine), XELIRI, XELOX, Xgeva (Denosumab), Xofigo (Radium 223 Dichloride), Xtandi (Enzalutamide), Yervoy (Ipilimumab), Yondelis (Trabectedin), Zaltrap (Ziv-Aflibercept), Zarxio (Filgrastim), Zejula (Niraparib Tosylate Monohydrate), Zelboraf (Vemurafenib), Zevalin (Ibritumomab Tiuxetan), Zinecard (Dexrazoxane Hydrochloride), Ziv-Aflibercept, Zofran (Ondansetron Hydrochloride), Zoladex (Goserelin Acetate), Zoledronic Acid, Zolinza (Vorinostat), Zometa (Zoledronic Acid), Zydelig (Idelalisib), Zykadia (Ceritinib), and/or Zytiga (Abiraterone Acetate). The treatment methods can include or further include checkpoint inhibitors including, but are not limited to antibodies that block PD-1 (such as, for example, Nivolumab (BMS-936558 or MDX1106), pembrolizumab, CT-011, MK-3475), PD-L1 (such as, for example, atezolizumab, avelumab, durvalumab, MDX-1105 (BMS-936559), MPDL3280A, or MSB0010718C), PD-L2 (such as, for example, rHlgM12B7), CTLA-4 (such as, for example, Ipilimumab (MDX-010), Tremelimumab (CP-675,206)), IDO, B7-H3 (such as, for example, MGA271, MGD009, omburtamab), B7-H4, B7-H3, T cell immunoreceptor with Ig and ITIM domains (TIGIT)(such as, for example BMS-986207, OMP-313M32, MK-7684, AB-154, ASP-8374, MTIG7192A, or PVSRIPO), CD96, B- and T-lymphocyte attenuator (BTLA), V-domain Ig suppressor of T cell activation (VISTA)(such as, for example, JNJ-61610588, CA-170), TIM3 (such as, for example, TSR-022, MBG453, Sym023, INCAGN2390, LY3321367, BMS-986258, SHR-1702, RO7121661), LAG-3 (such as, for example, BMS-986016, LAG525, MK-4280, REGN3767, TSR-033, B1754111, Sym022, FS118, MGD013, and Immutep)

D. EXAMPLES

[0154] The following examples are put forth so as to provide those of ordinary skill in the art with a complete disclosure and description of how the compounds, compositions, articles, devices and/or methods claimed herein are made and evaluated, and are intended to be purely exemplary and are not intended to limit the disclosure. Efforts

have been made to ensure accuracy with respect to numbers (e.g., amounts, temperature, etc.), but some errors and deviations should be accounted for. Unless indicated otherwise, parts are parts by weight, temperature is in ° C. or is at ambient temperature, and pressure is at or near atmospheric.

1. Example 1: Actionable Spontaneous Antibody Responses Antagonize Malignant Progression in Ovarian Carcinoma

a) Methods

(1) Human Samples

[0155] Human ovarian carcinoma tissues were procured under protocols approved by the Committee for the Protection of Human Subjects at Dartmouth-Hitchcock Medical Center (#17702), by the Institutional Review Board at Christiana Care Health System (#32214), and by Advarra Institutional Review Board (#0000971) and H. Lee Moffitt Cancer Center Scientific Review Committee (MCC #18974). Human endometrioma tissues were procured under a protocol approved by the Institutional Review Board at Ponce Research Institute (#1903009574). Informed consent was obtained from all subjects.

(2) Cell Lines and Culture Conditions

[0156] Human ovarian cancer cell lines including OVCAR3 (RRID: CVCL_0465), SKOV3 (RRID: CVCL_0532), and human endometrial stromal cells (HESC), highly invasive and immortalized with human telomerase reverse transcriptase (hTert), were obtained from ATCC. TOV21G (RRID: CVCL_3613), RMG-I (RRID: CVCL_1662), Caov3 (RRID: CVCL_0201), A2780 (RRID: CVCL_0134), OVCAR4 (RRID: CVCL_1627), OVCAR5 (RRID: CVCL_1628), OVCAR8 (RRID: CVCL_1629), Kuramochi (RRID: CVCL_1345) and BRCA OVCAR were obtained as a gift from Dr. Rugang Zhang at The Wistar Institute. Human endometrial epithelial cells (12Z, RRID: CVCL_QQ73) were obtained as part of a collaboration with Dr. Asgerally Fazleabas and Dr. Anna Starzinski-Powitz. All cell lines except RMG-I, HESC and 12Z were cultured in RPMI 1640 medium (Fisher Scientific) supplemented with 10% fetal bovine serum (FBS), penicillin (100 IU/mL), streptomycin (100 IU/mL), L-glutamine (2 mM), and sodium pyruvate (0.5 mM). RMG-I was cultured in Ham's F12 medium (Fisher Scientific) supplemented with 10% fetal bovine serum (FBS), penicillin (100 IU/mL), streptomycin (100 IU/mL), L-glutamine (2 mM), and sodium pyruvate (0.5 mM). 12Z was cultured in Dulbecco's Modified Eagle's Medium (DMEM)/F12 supplemented with 10% FBS. HESC was cultured in phenol-free DMEM supplemented with charcoal-treated 10% FBS and 1% Insulin-Transferrin-Selenium (ITS). All cell lines were routinely tested for *Mycoplasma* by PCR. Cells were used within 20 passages from thaw for in vitro experiments and 10 passages from thaw for in vivo experiments.

(3) Ovarian Cancer- and Endometriosis-Derived B Cell Immortalization, Antibody Purification, and Proteome Array

[0157] Cryopreserved single-cell suspensions of two stage IIIC ovarian clear cell ovarian carcinomas, two stage IIIC ovarian endometrioid carcinomas, and two endometriomas

were thawed and prepared, and CD19⁺ B cells isolated, activated, and immortalized, as previously described⁵. The conditioned medium from each was collected and concentrated using centrifugal filter units (Millipore Sigma Amicon, UFC900324). From the concentrated medium, human IgA and IgG were purified using immunoglobulin purification kits (LigaTrap, LT-146KIT and LT-095KIT) according to the manufacturer's protocols. To characterize the specificities of these antibodies, they were analyzed for reactivity against a proteome microarray that includes greater than 80% of the human proteome (HUPROTTM, CDI Laboratories, Mayaguez, Puerto Rico).

(4) Flow Cytometry

[0158] In order to determine which peptide would be best targeted by an antibody, the amino acid sequence for SDCBP was run through two epitope prediction tools (Bepipred Linear Epitope Prediction 2.0 and ABCPred) to determine predicted epitopes. Predicted targetable extracellular domains were chosen as the peptides for tetramer analysis. We tetramerized two biotinylated peptides (Genscript) contained in the extracellular domain of syndecan binding protein (SDCBP) using PE-labelled fluorescent streptavidin (BioLegend, 405203) and used flow cytometry to determine the percent of specific antigen-reactive immortalized B cells which are secreting antibodies against the target molecule from each of the six immortalized B cell lines. We then used fluorescence-activated cell sorting with EEEIRANVAVVSGAPL (SEQ ID NO: 1) peptide to sort endometrioid carcinoma-derived immortalized B cells specific for SDCBP from the pool of immortalized B cells.

(5) 10× Genomics single-cell V(D)J (BCR) sequencing and recombinant antibody production

[0159] Single-cell V(D)J B-cell receptor sequencing was performed by the Moffitt Cancer Center Molecular Genomics Core using the 10× Genomics Chromium system. 56 cells were encapsulated and sequenced. BCR reads sequenced by V(D)J assay were aligned to GRCh38 reference transcriptome using Cell Ranger VDJ (v.3.1.0, 10× Genomics). BCR heavy and light chains were assembled and annotated using Cell Ranger VDJ to determine clonotypes. Recombinant IgG4 antibodies were produced by Genscript. In brief, corresponding DNA sequences for the immunoglobulin heavy chain and light chain were synthesized and the complete sequence was subcloned into a pcDNA3.4 vector and expressed in HD 293F cells. IgG4 antibodies were eluted from cell culture supernatants. Molecular weight and purity were analyzed by SDS-PAGE and high-performance liquid chromatography.

(6) Real-Time Quantitative PCR

[0160] RNA was extracted from ovarian clear cell carcinoma, endometrioid carcinoma, and high-grade serous carcinoma tissues and cell lines using the RNEasy Plus Mini Kit (Qiagen) to quantify SDCBP expression. Total RNA were reverse transcribed using a high-capacity cDNA reverse transcription kit with RNase inhibitor (ThermoFisher, 4374966). Quantification of SDCBP was performed on the 7900HT Real-Time PCR system (Thermo Fisher Scientific) using SYBR Select Master Mix (Applied Biosystems, with forward primer: 5'-TCTCGAAGACTTGAAGGTAGACA-3'(SEQ ID NO: 2), and reverse primer: 5'-CGGCCACATTTCACGTATT-3'(SEQ ID NO: 3)).

Expression was normalized to levels of the endogenous reference control gene GAPDH (forward primer: 5'-CCTGCACCACCAACTGCTTA-3'(SEQ ID NO: 4); and reverse primer: 5'-AGTGATGGCATGGACTGTGGT-3'(SEQ ID NO: 5)).

(7) Western Blot Analysis

[0161] Proteins were extracted from ovarian clear cell carcinoma, endometrioid carcinoma, and high grade serous carcinoma tissues and cell lines, as well as an endometriosis cell line, and quantified. Proteins were loaded onto a 10% Bis-Tris polyacrylamide gel. Membranes were incubated with recombinant anti-SDCBP IgG4 antibodies described above (Genscript) or rabbit anti-human SDCBP (Sigma). After washing with TBST, the membranes were incubated with horseradish peroxidase-conjugated rabbit anti-human IgG (1:5000, Cat. Ab6759, Abcam, RRID:AB_955434). Horseradish peroxidase-conjugated anti-β-actin antibody (1:5000, Cat. 5125S, Cell Signaling Technology, RRID: AB_1903890) was used as a loading control. Images were captured using the BioRad ChemiDoc imaging system and GE Healthcare Amersham ECL Prime Western Blotting Detection Reagents (cat. 12316992, Fisher Scientific).

(8) In Vivo Models of High Grade Serous and Clear Cell Ovarian Cancer

[0162] All in vivo protocols were approved by the University of South Florida's Institutional Animal Care and Use Committee. Female NOD-SCID-gamma (NSG) mice, originally obtained from Jackson Laboratory, were maintained by the animal facility of H. Lee Moffitt Cancer Center and Research Institute. Mice were injected subcutaneously with 5×10⁶ RMG-I, TOV21G, or OVCAR3 cells in the right flank. Once tumor uptake was demonstrated, seven to nine days post-injection, mice were randomly divided into two treatment groups of five mice each: irrelevant IgG4 control and anti-SDCBP IgG4 treatment. Twice weekly, tumor volume was measured and IgG4 control and anti-SDCBP IgG4 treatment were administered by intratumoral or peritumoral injection (100 μg/100 μL). Tumor volume was calculated as (L×W²)/2, in which L is length and W is width. Once any group became moribund, all mice were euthanized and tumor weight was recorded. Tumor specimens were formalin-fixed and paraffin-embedded or mechanically dissociated into single-cell suspensions.

(9) Statistical Analysis and Reproducibility

[0163] All experiments were repeated at least twice with similar results. The Shapiro-Wilk test was applied to determine whether data were normally distributed. Wilcoxon matched-pairs signed rank tests and unpaired t tests were applied as indicated. Two-tailed analyses were performed. Error bars represent standard error of the mean. A p value less than 0.05 was considered statistically significant. Statistical analyses were performed using GraphPad Prism 9.0 (GraphPad Prism, RRID:SCR_002798).

b) Results

(1) Tumor Infiltrating B Cells in Clear Cell Ovarian Carcinoma, Endometrioid Ovarian Carcinoma and B Cells in the Microenvironment of Endometriosis Share Common Antigen Targets

[0164] To identify antibodies produced by B cells recognizing possible shared targets in clear cell ovarian carci-

noma, endometrioid ovarian carcinoma, and endometriosis, we analyzed viable single-cell suspensions from two cryo-preserved samples of each type. B cells were isolated, activated, and immortalized using Epstein-Barr virus. These six immortalized B cell pools were found to secrete IgG and IgA at titers in the 0.7-37 mg/mL range. Using HUPROT™ proteome arrays containing >80% of the human proteome, IgG and IgA tumor reactivities were decoded (FIG. 1A). Greater than 200 targets were identified for each sample, for both IgA and IgG antibodies in independent analyses.

[0165] There were nine molecules that met the following criteria: 1) They were either secreted or included an extracellular domain, and were therefore accessible to antibodies in live cells; 2) they were recognized by both IgA and IgG; and 3) reactivity was shared in every endometrioma and carcinoma sample (FIG. 1B). Among these, we focused on SDCBP, a transmembrane molecule that links syndecan-mediated signaling to the cytoskeleton. SDCBP regulates TGF- β 1-induced Smad activation and EMT by inhibiting caveolin-mediated TGF- β type I receptor internalization, and has been demonstrated to drive proliferation, migration, invasion, and angiogenesis. SDCBP has been associated with unfavorable prognosis in multiple solid malignancies, including breast and colorectal cancer. Furthermore, SDCBP has been reported as a therapeutic target for cancer metastases, as well as cancer sternness and chemoresistance.

[0166] Because antibodies targeting SDCBP are being spontaneously produced in the endometriosis and ovarian cancer microenvironment, we tetramerized two different biotinylated 16-20mer peptides contained in the extracellular domain of SDCBP using fluorescent streptavidin and used flow cytometry to determine the percent of specific antigen-reactive immortalized B cells which are secreting antibodies against SDCBP from each of the six immortalized B cell lines. Based on these results, we used fluorescent activated cell sorting to sort endometrioid ovarian cancer-derived B cells specific for SDCBP and performed single-cell B cell receptor sequencing on the sorted population of B cells (FIG. 2). Bioinformatic analysis of the B cell receptor sequencing determined the sequence of the heavy chain and light chain of the most common B cell receptor, identified in 96% of these cells (FIG. 2). We then produced a recombinant antibody targeting SDCBP using these heavy chain and light chain sequences on an IgG4 backbone. IgG4 was specifically selected to avoid antibody-dependent killing of normal cells that also express SDCBP via antibody-dependent cell-mediated cytotoxicity or antibody-dependent cellular phagocytosis.

(2) SDCBP is Expressed in the Majority of Clear Cell, Endometrioid, and High Grade Serous Ovarian Carcinomas

[0167] The Cancer Genome Atlas RNA sequencing data were queried, demonstrating SDCBP mRNA expression in all tested cancer types, supporting its potential as a therapeutic target for patients with a diverse range of histologies (FIG. 3A, FIG. 6). RT-qPCR was performed to assess the baseline prevalence of SDCBP mRNA expression in ovarian cancer cell lines and tumor tissues. 26/29 (90%) of the cell lines and tissue samples express SDCBP mRNA—and all of the clear cell (5/5) and endometrioid (7/7) cell lines and tissue samples express SDCBP (FIG. 3B). Tumor-derived, recombinant anti-SDCBP IgG4 recognizes recombinant SDCBP in Western blot analysis. Western blot demonstrated SDCBP protein expression in ovarian cancer cell lines and

tumor samples tested, including clear cell, endometrioid, and high grade serous histologies (FIG. 3C, FIG. 7), as well as expression in endometriosis (FIG. 3D).

(3) SDCBP-Specific IgG4 Abrogates Ovarian Cancer Progression

[0168] To investigate the antitumor effects of a recombinant anti-SDCBP IgG4 antibody in vivo, we created a high grade serous ovarian carcinoma mouse model using subcutaneous injection of OVCAR3 cells (FIG. 4A). Mice treated with the anti-SDCBP IgG4 antibody demonstrated significantly reduced tumor growth and substantially reduced tumor weight compared to those treated with the control irrelevant IgG4 antibody ($p=0.004$ and $p=0.05$, respectively, FIG. 4B-D). To determine whether this can be extrapolated to other ovarian cancer histologies, specifically clear cell carcinoma—one of the endometriosis-associated ovarian cancer types, we next evaluated the anti-SDCBP antibody in an RMG-I tumor model. In this model, we also observed significantly reduced tumor growth and lower tumor weight in the treatment arm compared to the control arm ($p<0.05$ and $p=0.34$, respectively, FIG. 5A-B). To confirm these findings, we performed the same experiments in a second clear cell line, TOV21G, which yielded similar results (FIG. 5C-D).

c) Discussion

[0169] A novel anti-SDCBP IgG4 antibody has demonstrated preclinical anti-tumor efficacy in HGSOC and CCC, with the possibility of use in EC and other tumor types given the broad expression of SDCBP among tumors. As HGSOC is the most common histologic type of ovarian cancer, and as CCC is relatively chemotherapy-resistant and associated with increased risk of poor outcomes, these two disease types represent an area of high unmet need for novel therapeutic strategies. This study establishes the potential of this technique in identifying novel therapeutic targets for CCC and verifies the utility of the technique for HGSOC, in which we had previously identified SDCBP as a target of tumor-infiltrating, IgG-producing B cells in six tumor samples. This study also highlights the importance of antibody responses in different histologic subtypes of ovarian cancer and supports that intratumoral B cells, through the spontaneous production of antibodies, can exert a protective role against malignant progression. SDCBP is present in small extracellular vesicles; we did not have enough serum samples from the patients analyzed to perform ELISAs, but it is theoretically possible that reactive antibodies can be detected in circulation. The concordance of antibodies being spontaneously produced in the microenvironment of endometriosis and endometriosis-related ovarian cancers raises the question of whether antibodies such as those targeting SDCBP can prevent or delay progression from endometriosis to EAOCs—in addition to delaying tumor progression in established EAOCs.

[0170] In patients with ovarian cancer, high SDCBP expression is associated with a slight trend towards worse OS ($p=0.37$). The issue, however, is that the level of expression of SDCBP is very high in virtually all included ovarian cancers in this analysis as seen in FIG. 3A, while there is more dispersion in the level of expression of SDCBP in other gynecologic and other cancer types, allowing a better comparison between high- and low-expressing tumors. Of

note, high SDCBP expression is associated with statistically significant worse OS in breast, cervical, and endometrial cancers, among others FIG. 8) (Cancer Gene Prognosis Atlas), indicating multiple avenues for further research in other cancer types—gynecologic and otherwise.

d) Conclusion

[0171] An α -SDCBP IgG4 has demonstrated anti-tumor efficacy in SDCBP⁺ CCC and HGSOE, and SDCBP-targeted therapy for endometriosis and associated malignant conditions, as well as HGSOE, warrants further investigation.

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F. SEQUENCES

SEQ ID NO: 1
EEEIRANVAVVSGAPL

SEQ ID NO: 2 SDCBP forward primer:
TCTCGAAGACTTGAAGGTAGACA

SEQ ID NO: 3 SDCBP reverse primer
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-continued

SEQ ID NO: 4 GAPDH forward primer
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SEQ ID NO: 5 GAPDH reverse primer
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SEQ ID NO: 6 anti-SDCBP variable heavy chain
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GYTFTSYD

SEQ ID NO: 8 anti-SDCBP variable heavy chain CDR2
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SEQ ID NO: 9 anti-SDCBP variable heavy chain CDR3
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GAS

SEQ ID NO: 13 anti-SDCBP variable light chain CDR3
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SEQ ID NO: 14 IgG1 VL CDR3
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SEQ ID NO: 15 IgG1 VH1-8 CDR3
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SEQ ID NO: 17 IgG1 IgHV1-2 VH CDR3
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SEQUENCE LISTING

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 organism = synthetic construct

SEQUENCE: 8
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organism = synthetic construct		
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GPGTKVDIK	129	

1. An anti-syndecan binding protein (SDCBP) binding molecule comprising a heavy chain variable domain complementary determining region (CDR) 1 (CDR1), CDR2, and CDR3 as set forth in SEQ ID NO: 7, SEQ ID NO: 8, and SEQ ID NO: 9, respectively; SEQ ID NO: 7, SEQ ID NO: 8, and SEQ ID NO: 15, respectively; SEQ ID NO: 7, SEQ ID NO: 8, and SEQ ID NO: 16, respectively; or SEQ ID NO: 7, SEQ ID NO: 8, and SEQ ID NO: 17, respectively.

2. The anti-SDCBP binding molecule of claim 1, wherein the heavy chain variable domain comprises the amino acid sequence as set forth in SEQ ID NO: 6 or SEQ ID NO: 18.

3. The anti-SDCBP binding molecule of claim 1, wherein the anti-SDCBP binding molecule further comprises a light chain variable domain complementary determining region (CDR) 1 (CDR1), CDR2, and CDR3 as set forth in SEQ ID NO: 11, SEQ ID NO: 12, and SEQ ID NO: 13, respectively or SEQ ID NO: 11, SEQ ID NO: 12, and SEQ ID NO: 14, respectively.

4. The anti-SDCBP binding molecule of claim 1, wherein the light chain variable domain comprises the amino acid sequence as set forth in SEQ ID NO: 10 or SEQ ID NO: 19.

5. The anti-SDCBP binding molecule of claim 1, wherein binding molecule comprises an antibody, nanobody, or single chain variable fragment (scFv).

6. A method of treating a endometriosis or cancer in a subject comprising administering to the subject the anti-SDCBP binding molecule of claim 1.

7. A method of treating a endometriosis or cancer in a subject comprising administering to the subject an anti-syndecan binding protein (SDCBP) binding molecule com-

prising a heavy chain variable domain complementary determining region (CDR) 1 (CDR1), CDR2, and CDR3 as set forth in SEQ ID NO: 7, SEQ ID NO: 8, and SEQ ID NO: 9, respectively; SEQ ID NO: 7, SEQ ID NO: 8, and SEQ ID NO: 15, respectively; SEQ ID NO: 7, SEQ ID NO: 8, and SEQ ID NO: 16, respectively; or SEQ ID NO: 7, SEQ ID NO: 8, and SEQ ID NO: 17, respectively.

8. The method of treating a cancer of claim 7, wherein the heavy chain domain comprises the amino acid sequence as set forth in SEQ ID NO: 6 or SEQ ID NO: 18.

9. The method of treating a cancer of claim 7, wherein the anti-SDCBP binding molecule further comprises a light chain variable domain complementary determining region (CDR) 1 (CDR1), CDR2, and CDR3 as set forth in SEQ ID NO: 11, SEQ ID NO: 12, and SEQ ID NO: 13, respectively or SEQ ID NO: 11, SEQ ID NO: 12, and SEQ ID NO: 14, respectively.

10. The method of treating a cancer of claim 7, wherein the heavy chain domain comprises the amino acid sequence as set forth in SEQ ID NO: 10 or SEQ ID NO: 19.

11. The method of treating a cancer of claim 7, wherein binding molecule comprises an antibody, nanobody, or single chain variable fragment (scFv).

12. The method of treating a cancer of claim 6, wherein the cancer comprises an ovarian cancer.

13. The method of treating a cancer of claim 12, wherein the ovarian cancer comprises clear cell carcinoma (CCC), endometrioid carcinoma (EC), high-grade serous ovarian carcinoma (HGSOC), or endometriosis-associated ovarian cancer (EOAC)).

* * * * *